FUSION MENU · FOR SERVER OWNERS

Server Owner's Guide

Installing, becoming the owner, the admin panel, roles and permissions, every feature, adding content, running QBCore scripts, keys and commands, backups and updates, and troubleshooting.

Fusion Menu v1.0.1 build 230

Everything here is set in game: press F10 for the admin panel.

Latest version of this guide: github.com/Marine1108/fusion-menu-server-kit

For your staff: the Fusion Management Guide. For your players: the Fusion Player Guide.

Support: discord.gg/kVRqwDPsnJ

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Fusion Menu is a FiveM framework: a set of resources (`fusion-core` , `fusion-inventory` , `fusion-vehicles` and the other `fusion-…` modules) that you add to your FiveM server. It works for **freeroam and roleplay**, is **compatible with QBCore scripts without modification**, and is **configured entirely in game**. Settings, permissions, locations, jobs, items, vehicles, prices and every feature toggle live in the admin panel. The only files you ever add are pictures, music and new resources.

What you get: the Fusion Menu resources come from the Cfx.re portal after buying. Everything else in this guide (the installers, the `server.cfg` template, the txAdmin recipe and the QBCore compatibility bridges in `[compat]`) is the **free server kit**: optional tools that set up a FiveM server for Fusion Menu, or add Fusion Menu to the server you already have.

As a PDF: this guide is also the Fusion Server Owner's Guide (PDF), which is in the server kit too.

For your staff: the Fusion Management Guide (PDF) is the day-to-day handbook: reports, moderation, the anticheat, managing players and staff, departments, upkeep, updates and backups.

For your players: the Fusion Player Guide (PDF) explains the keys, the menu, the phone, vehicles, homes, jobs and how to get help. Share it on your Discord or website; it is also in the server kit (`docs/player-guide`).

1. Folder layout

FusionMenu/	your server data folder
├─ install-windows.bat	Windows installer (double-click)
├─ install-linux.sh	Linux / VPS installer
├─ update-fusion.bat / .sh	update Fusion Menu without losing your changes
├─ install-script.bat / .sh	install or upgrade QBCore scripts, vehicles, EUP, maps
├─ updates/	put Fusion update zips here
├─ backups/	automatic backups made by the two tools above
├─ recipe.yaml	txAdmin one-click recipe
├─ install/	installer files and the server.cfg template
├─ server.cfg	written by the installer
└─ resources/	
├─ [cfx-default]	standard FiveM resources (chat, spawnmanager…)
├─ [fusion]	the framework, do not remove anything here
│ ├─ oxmysql	database driver
│ ├─ fusion-core	settings, permissions, characters, economy, jobs, admin panel, UI
│ ├─ fusion-anticheat	anticheat: detections, honeypots, entity & explosion protection
│ ├─ fusion-loadscreen	loading screen
│ ├─ fusion-target	third-eye targeting (hold Left Alt)
│ ├─ fusion-inventory	inventory, stashes, trunks, drops
│ ├─ fusion-vehicles	spawner, vehicle options, garages, keys
│ ├─ fusion-emergency	siren controller, police radar and plate reader
│ ├─ fusion-phone	phone: calls, texts, contacts, 911, bank, weather, web, emergency / AMBER alerts
│ ├─ fusion-voice	proximity voice, radio and phone call audio (pma-voice compatible)
│ ├─ fusion-postalmap	the illustrated pause map with postal codes and place names (F10 › Postal Map)
│ ├─ fusion-vinewood	the hillside sign: your own text, colors and night glow (F10 › Vinewood Sign)
│ ├─ fusion-housing	apartments and houses with real interiors, keys and doorbells (F10 › Housing)
│ ├─ fusion-updater	automatic Fusion updates: signed, checksummed, scheduled, rolled back if needed (F10 › Updates)
│ ├─ fusion-weapons	weapon spawner, loadouts, weapon wheel
│ ├─ fusion-appearance	character creator, clothing, EUP uniforms
│ ├─ fusion-stores	shops
│ ├─ fusion-fuel	fuel and gas stations
│ ├─ fusion-hospital	death, respawning, EMS
│ ├─ fusion-jail	prison
│ ├─ fusion-weather	weather and time sync
│ └─ [compat]	qb-core, qb-menu, qb-input, progressbar: free QBCore compatibility bridges (server kit)
├─ [standalone]	
│ └─ screenshot-basic	screenshots for other scripts (CCTV, evidence, MDTs, phone cameras)
├─ [qb]	put QBCore scripts here
└─ [addons]	put vehicles, EUP, peds, maps here

Everything in `[qb]` and `[addons]` starts automatically, so you normally never edit `server.cfg` to add content. The only exception is scripts that must start in a particular order (section 8).

2. Installing and first start

You need: a FiveM server license key (free at portal.cfx.re › Server Registration Keys). The installers download and set up everything else: the database (MariaDB), FXServer and txAdmin. If you set things up by hand: MySQL or MariaDB, and FXServer build 12913 or newer.

Getting Fusion Menu: after buying, sign in at portal.cfx.re with the same Cfx.re account and open **Granted Assets**. Fusion Menu is protected by Cfx.re, so only you can download it, and it only runs on servers whose license key belongs to that account. Download the zip there; the installers below find it and put it in the right place for you. Everything else (the free server kit, database, FXServer) is downloaded by the installer.

Pick the one that matches where your server runs.

Option A: Your own Windows PC (one click)

1. Download Fusion Menu from portal.cfx.re › Granted Assets (leave it in your **Downloads** folder).
2. Download **FusionMenu-Setup.bat** and double-click it. Click **Yes** when Windows asks for permission. (If Windows says *"Windows protected your PC"*, click **More info › Run anyway**.)
3. Answer the questions (press Enter for the suggested answer). The setup:
   - downloads the server kit into `C:\FusionServer` (or a folder you choose) and unpacks your Fusion Menu zip into it,
   - **sets up the database so it starts by itself with Windows.** You never have to open XAMPP:
     - no database yet → it installs **MariaDB** as a background Windows service,
     - XAMPP installed → it turns XAMPP's MySQL into a Windows service,
     - your own MySQL → it just asks for the login,
   - creates the database and a login just for Fusion (saved in `database-login.txt`, keep it private),
   - asks for your license key, server name, players, port and (optionally) your identifier, and writes `server.cfg`,
   - downloads the recommended **FXServer** build to `C:\FXServer`,
   - opens the port in the Windows firewall and, if you like, starts the server when you log in to Windows,
   - puts **Start Fusion Server** and **Fusion txAdmin** shortcuts on your desktop.
4. Double-click **Start Fusion Server**. That's it.

Already have the kit folder? Double-click `install-windows.bat` inside it instead; it does the same from step 3.

Option B: Your own Linux VPS (one command, Ubuntu / Debian)

1. Upload the Fusion Menu zip from portal.cfx.re to the VPS (WinSCP or FileZilla, e.g. into `/root`).
2. Run:

```
curl -fsSL https://raw.githubusercontent.com/Marine1108/fusion-menu-server-kit/main/setup/get-fusion.sh | sudo bash
```

3. Answer the questions. It downloads the server kit to `/home/fivem/fusion`, finds and unpacks your Fusion Menu zip, installs **MariaDB** with a random password, writes `server.cfg`, downloads **FXServer** to `/opt/fxserver`, creates a `fusion` **systemd service** (starts on boot, with or without the txAdmin web panel) and opens the firewall ports.
4. **txAdmin mode:** open `http://YOUR_SERVER_IP:40120`, enter the PIN from `journalctl -u fusion -f`, choose **Existing server data** with the Fusion folder and `server.cfg`, turn OneSync on and start. **Direct mode:** the server is already running (`systemctl start|stop|restart fusion`, logs: `journalctl -u fusion -f`).

Option C: A game server host (ZAP-Hosting and others with txAdmin)

1. In the host's panel, open **txAdmin** and start the setup. Choose **Popular Template › Remote URL Template** and paste:

   ```
   https://raw.githubusercontent.com/Marine1108/fusion-menu-server-kit/main/recipe.yaml
   ```

2. Enter your license key. For the database, use the details from your host's panel (most hosts fill them in for you). txAdmin downloads the server kit, creates the database, writes `server.cfg` and makes your txAdmin account the owner. OneSync is turned on for you.
3. Upload Fusion Menu: unzip the file from portal.cfx.re on your PC, then upload its folders into `resources/[fusion]` with the host's **File Manager** or FTP (next to `oxmysql`).
4. Start the server in txAdmin.

Option D: Your own server with txAdmin's recipe

Prefer txAdmin's recipe on your own PC or VPS instead of Option A or B? txAdmin's database step needs a database login it can sign in to with a password, which a fresh machine doesn't have. Set one up first:

- **Linux VPS (Ubuntu / Debian):** run

  ```
  curl -fsSL https://raw.githubusercontent.com/Marine1108/fusion-menu-server-kit/main/setup/prepare-database.sh | sudo bash
  ```

  It installs MariaDB if needed, creates the login `txadmin` with a random password, and prints what to type in txAdmin's database step (also saved in `~/fusion-database-login.txt`).
- **Windows:** install a database that starts with Windows, e.g. run **FusionMenu-Setup.bat** once (it installs MariaDB and saves the login in `database-login.txt`), or use XAMPP's MySQL (user `root`, no password). Using the setup? Then you don't need the recipe at all, see below.

Then follow Option C, steps 1–4, using that login on the database step.

> **Already ran the Windows setup or the Linux installer?** Don't use the recipe: everything is set up. In txAdmin choose **Existing Server Data** and point it at the server folder (the installer prints the exact paths; on Linux with txAdmin it's `/opt/fxserver/txData/fusion`). Running the recipe as well would create a second, separate server.

Players connect on port **30120** (TCP and UDP) unless you chose another. txAdmin uses **40120**. If friends outside your home can't join a server on your own PC, forward port 30120 on your router.

First start

The console ends with:

```
[Fusion] Ready - open the admin panel in game with /admin (or F10)
```

Join the server, and press **F10** once you are the owner (see below).

What players see when joining

Loading screen → **player guide** (first visit only) → character selection → **spawn selector** → in the city. New characters get the appearance editor right after spawning.

3. Becoming the owner and adding staff

Owners always have every permission. You become one in any of these ways:

- **Claim code (easiest):** if no owner is set, the server console shows `Join the server and type: /claimowner ABC123`. Type that in the in-game chat. The code works once and changes every restart until it's used.
- **Installer / txAdmin:** the identifier you enter (or your txAdmin account) is added to `group.admin`, and everyone in `group.admin` is an owner.
- **Server console:** `fusion_role <server id or identifier> superadmin`.

Everyone else is managed in game.

To add staff: F10 › **Roles & Permissions** › pick a role › **Members** › **Add online player** (or **Add by identifier** for someone offline: `license:…`, `discord:…`, `fivem:…`, `steam:…`).

From the server console:

```
fusion_role <server id or identifier> <role>
fusion_role license:abc123… admin
```

4. The admin panel

Open with **F10** or `/admin`. Each tab only appears for staff with the matching permission.

TAB	WHAT YOU CAN DO
Dashboard	Players online, staff online, characters, bans, uptime, recent activity, quick actions (noclip, teleport to waypoint, announcement, delete empty vehicles)
Players	Search online players. Open one to: go to / bring, spectate, revive, heal, freeze, open inventory, message, send to character select, kill, warn, kick, ban, give/take/set money, set job and gang, give/remove items, clear inventory, add/remove roles, see identifiers and warnings
Characters	Search every character (online or offline). Edit names, money and job, or delete a character
Reports	Player reports and admin requests: claim, go to, bring, message, close with a note. A red badge shows how many are open
Bans	Search bans, unban, ban an identifier of someone who is offline
Roles & Permissions	Create roles, set allow/deny rules, manage members, add permission nodes for other scripts (section 5)
Settings	One page per module, plus every location in the server. Export / Import backs up all settings
Jobs / Gangs	Create and edit jobs and gangs: grades, paychecks, boss grades, duty behavior, whitelisting
Items	Create and edit items: label, weight, picture, description, usable, unique, and what happens when used (hunger, thirst, health, armor, stress, animation)
Vehicles	The vehicle catalog (every vanilla and addon vehicle): names, brands, prices, class, picture, hidden, permission or job restrictions
Handling	In-game handling editor for addon vehicles. Saves straight into the vehicle's `handling.meta`
Sirens & Carcols	Finds siren, light-settings and modkit IDs used by more than one vehicle pack (wrong sirens, lightbars or missing mods) and gives the clashing packs free IDs, with backups and undo
Anticheat	Every detection with player, details and the action taken. Filter by detection, search, open the player, ban them, delete false positives, clear old detections
Logs	Searchable history of admin actions, money, items, connections, jobs, settings and permission changes
Server	Announcements, clean-up tools (empty vehicles, NPCs, props: within a distance of you, or the entire server), and start / stop / restart for any resource

Using your position in settings

Every location setting (garages, stores, stashes, hospitals, banks, spawns, teleports...) has a **Here** button that fills in your current position and heading. When you add a new entry to a list, every position in it starts where you stand. Stand where you want the thing, click **Add**, give it a name, done. Changes to locations (garages, impound lots, stores, dealerships, refuel points, blips...) show for every player straight away, with no restart.

Lists to pick from (items, weapons, vehicles, jobs, store catalogs…) stay up to date while F10 is open: an item, weapon, vehicle or catalog added in game appears in every list at once.

Staff hierarchy

Each role has a **priority**. Staff can only kick, ban, warn, freeze, kill or log out players whose priority is lower than their own, and can only hand out roles below their own. Owners are exempt.

5. Roles and permissions

Default roles

ROLE	PRIORITY	INHERITS	HAS
Player (`user`)	0	—	Everyday features. **Every player has this role.**
Moderator	50	Player	Admin panel, kick, warn, freeze, revive, heal, teleport, spectate, noclip, logs…
Admin	100	Moderator	`admin.*` (everything admin)
Owner (`superadmin`)	1000	Admin	`*` (everything). Cannot be deleted

You can rename, recolor, re-prioritize and re-wire all of these, and create as many new roles as you like (e.g. *LSPD Command*, *Trusted*, *Donator*).

How a permission is decided

On the **Permissions** tab, every permission has three buttons: **Allow**, **Inherit** and **Deny**.

- **Inherit** means the role doesn't decide it; the role it inherits from does. The gray/green text shows the current result and which role it came from.
- **Wildcards:** `admin.*` covers every node starting with `admin.` , and `*` covers everything. Add them under *Wildcard & custom rules*.
- **The most specific rule wins.** `vehicles.godmode = Deny` beats `vehicles.* = Allow` . A deny beats an allow when they are equally specific.
- A player with several roles gets the combination of all of them.

Permissions for other scripts (ACE)

Fusion mirrors every role to FiveM's ACE system as `group.fusion_<role>` , so standalone scripts that use `IsPlayerAceAllowed` or restricted commands follow your in-game roles.

THE SCRIPT CHECKS…	DO THIS
An ACE permission such as `cctv.view`	Open the role › **Wildcard & custom rules** › type `cctv.view` › **Allow** › **Add rule**. Several at once: `cctv.view, cctv.record, cctv.playback` ; all of a script's permissions: `cctv.*`
A restricted command such as `/carwash`	Add the rule `command.carwash` and allow it
QBCore permission levels (`admin` , `god` , `mod`)	Allow `qbcore.admin` , `qbcore.god` or `qbcore.mod` on a role
A job ("police only")	Nothing; jobs are edited in **Jobs**

> Scripts that keep their own admin list inside their `config.lua` (identifiers, Discord roles) cannot be controlled from Fusion. That list still lives in the script.

Useful permission nodes

AREA	NODES
General	`menu.open`, `server.whitelisted`, `server.maintenance`, `characters.extraslots`, `guide.open`, `reports.create`
Player menu	`player.menu`, `player.godmode`, `player.invisible`, `player.heal`, `player.stamina`, `player.fastrun`, `player.superjump`, `player.noragdoll`, `player.walkstyle`, `player.scenarios`
Misc	`misc.tpwaypoint`, `misc.teleports`, `misc.coords`, `misc.nightvision`, `misc.thermal`
Admin	`admin.panel`, `admin.players`, `admin.kick`, `admin.ban`, `admin.unban`, `admin.warn`, `admin.money`, `admin.items`, `admin.setjob`, `admin.roles`, `admin.permissions`, `admin.settings`, `admin.jobs`, `admin.registry`, `admin.logs`, `admin.resources`, `admin.noclip`, `admin.spectate`, `admin.blips`, `admin.names`, `admin.reports`, `admin.handling`, `admin.carcols`, `admin.scriptconfig`, `admin.anticheat`
Vehicles	`vehicles.spawn`, `vehicles.spawn.byname`, `vehicles.spawn.free`, `vehicles.class.<class>` (e.g. `vehicles.class.emergency`), `vehicles.addons`, `vehicles.extras`, `vehicles.livery`, `vehicles.mods`, `vehicles.godmode`, `vehicles.performance`, `vehicles.garage`, `vehicles.keys.all`
Weapons	`weapons.spawn`, `weapons.category.<category>` (e.g. `weapons.category.heavy`), `weapons.all`, `weapons.attachments`, `weapons.unlimitedammo`, `weapons.loadouts`
Appearance	`appearance.anywhere`, `appearance.face`, `appearance.peds`, `appearance.uniforms`, `appearance.uniforms.manage`, `appearance.free`
Anticheat	`anticheat.bypass` (never flagged; admins by default), `anticheat.manage` (delete detections), `admin.anticheat` (Anticheat tab and staff alerts)
Emergency	`emergency.siren`, `emergency.radar`, `emergency.bolo`, `emergency.pit`
Phone	`phone.use`, `phone.amber`, `admin.alerts`
Updates	`admin.updates` (F10 › Updates: install, schedule, skip and undo Fusion updates; owners by default)
Streaming & video	`media.streamer`, `media.videomode`, `media.freecam`, `media.freecam.unlimited`, `media.record`, `media.rockstar`
Other	`inventory.search`, `stores.free`, `fuel.free`, `hospital.revive`, `jail.use`, `weather.change`, `weather.events`, `time.change`

The **Permissions** tab lists every node with a description, so you don't need to memorize these.

6. Turning features on and off

Replacing a Fusion module with another script

Every Fusion module can be swapped for a script you prefer. In **Settings › Modules**, turn off *Use Fusion's phone* (or inventory, voice, targeting, fuel, garages, hospital, jail, door locks, stores, weapons, appearance, sirens/radar, weather, loading screen or anticheat), then install the replacement in `resources/[qb]`. Fusion stops its own module straight away and after every restart, and the rest of Fusion keeps working without it. Turn the switch back on to return to Fusion's version. `install-script` refuses scripts Fusion already replaces until you turn Fusion's version off and run it with `-Force` (`--force` on Linux).

Every module has a page in **F10 › Settings**, and the first switch on it turns the whole module on or off. Changes apply instantly for everyone.

SETTINGS PAGE	MAIN SWITCH	NOTES
Economy	**Enable economy**	Off = everything free, money hidden, no paychecks / banks / giving cash
Jobs & Gangs	Enable job features	
Vehicles	Enable vehicle module	Spawner, saved vehicles, keys and seatbelt also have their own switches
Garages & Impound	Enable garages	
Emergency Vehicles	Enable emergency vehicle features	Siren controller and radar each have their own switch
Weapons	Enable weapon module	**Mode:** free (freeroam) or inventory items (RP)
Appearance & EUP	Enable appearance module	*Editor available anywhere* vs stores only
Inventory	Enable inventory	Hotbar, giving and dropping have their own switches
Stores	Enable stores	
Fuel & Gas Stations	Enable fuel	Off = vehicles never run out
Hospital & Death	Enable hospital module	**Death mode:** respawn (freeroam) or downed/bleed-out (RP)
Jail	Enable jail	
Door Locks	Enable door locks	The doors themselves are listed on the same page
Weather & Time	Sync weather & time	
Targeting	Enable targeting	
Player Guide	Show the guide to new players	
Map Blips	Show these blips	Police and fire stations and custom map icons

Anticheat	Enable anticheat	Each detection can also be set to off / log / alert / kick / ban
Reports & Help	Enable reports	Player reports, admin requests and questions can each be turned off
Gameplay	—	Hunger & thirst, PvP, wanted level, NPC/traffic density, HUD parts, player blips and names

To hide a feature from some players only, take away its permission instead (e.g. remove `player.godmode` from Player).

> Stopping a resource on the **Server** tab only lasts until the next restart. Use the switches above for permanent changes.

7. Feature guides

Server, branding and interface

- **Settings › Server:** server name, server-list description and tags (update live), Discord link, autosave interval, **whitelist** (only `server.whitelisted` can join), **maintenance mode** (only `server.maintenance` can join), join/leave messages.
- **Settings › Interface:** menu title, subtitle and logo, accent colors, panel color and opacity, menu position (left/right/center) and size, notification position, money pop-ups, HUD, menu sounds, and whether players can walk/drive with the menu open.
- **Logo:** the menu shows the Fusion logo until you set your own. Click **Upload** next to *Logo* in **Settings › Interface** and pick a picture (PNG, JPG, WebP, GIF or SVG): it shows for everyone straight away, no restart. You can also type a web link, or a file name from `resources/[fusion]/fusion-core/html/img/` (a file copied there only reaches players after `ensure fusion-core`). To show no logo at all, click **Clear** and turn off *Show the Fusion logo when no logo is set*. Every menu ends with *Fusion Menu by Fusion Platform Technologies*.

HUD and vehicle HUD

- Players see their server ID, cash, bank, job and status (noclip, jail…). In a vehicle they get the **vehicle HUD**: speed, gear, RPM, fuel (or battery), engine warning, seatbelt, lock status, street and compass.
- **Every player can move their own HUD:** *Menu › Misc Settings › Edit HUD Layout* or `/hud`. Drag anything (cash, bank, job, status, vehicle HUD, notifications, progress bar, prompts) where you want it, and right-click to hide or show it. **Reset** restores the defaults. Layouts are saved per player.
- **Settings › Interface › HUD:** turn the HUD, vehicle HUD and street/compass on or off, and decide whether players may move the HUD. Speed units (MPH / KM/H) are in **Settings › Vehicles**.
- **Speedometer styles:** *Classic card*, *Analog dial* (needle gauge with an RPM ring), *Digital (LCD)*, *Sport* (shift-light ribbon) and *Minimal*. The server picks one in **Settings › Interface › HUD › Speedometer style**; with *Players can pick their own speedometer* on, each player chooses theirs in *Menu › Misc Settings › Speedometer*.
- **Flight instruments:** in a **plane or helicopter** the speedometer becomes a flight display, whatever style is picked: airspeed (knots) with an artificial horizon, then **ground speed** in its own section, in the server's speed unit (MPH or KM/H) and in knots, so air units can call out a pursuit's speed. Ground speed is the speed over the ground, so climbing or descending doesn't change it. Below that: altitude and height above ground (feet), climb rate, heading, landing gear (planes), fuel, engine, and the postal and street below you. Switch it off in **Settings › Interface › HUD › Flight instruments in planes and helicopters**.
- **Terrain warning:** in a plane or helicopter, *SINK RATE*, *PULL UP* or *TERRAIN AHEAD* flashes on the flight display with a warning tone when you're about to hit the ground, a hill or a building (it looks about 6 seconds ahead along your flight path). It stays quiet on the ground, while hovering and on normal landings (gear down, or a slow final descent). The server can switch it off (**Settings › Interface › HUD › Terrain warning in planes and helicopters**) and each player can too (*Menu › Misc Settings › Terrain warning*).

- **Windshield HUD:** driving or riding in **first person** (press V until you're in the cockpit), the speedometer is replaced by a see-through display low on the windshield: speed, gear, RPM bar, turn signals, high beams, compass and street, fuel, and blinking *FASTEN SEATBELT* / *ENGINE DAMAGE* warnings. It works in every vehicle. In a **plane or helicopter** it becomes a flight display: airspeed (knots), altitude and climb rate (feet), ground speed, heading, an artificial horizon that follows your pitch and roll, radar altitude near the ground, landing gear, and a *TOO LOW · GEAR* warning. The server can switch it off (**Settings › Interface › HUD**) and each player can too (*Menu › Misc Settings › Windshield HUD*). Players can move it with `/hud`.

Streaming & video (M › Streaming & Video)

Streamer mode hides what each player chooses from their own screen, so it never shows on their stream or in their recordings: the chat box, names and IDs above players, names and IDs in menus, their own server ID, cash and bank, job, street / postal / compass, the minimap, phone numbers, the voice and radio channel, and notifications. *What To Hide* has a checkbox for each. Bind *Fusion: streamer mode, show everything while held* in the key bindings to see it all for a moment without it staying on screen. Also `/streamer`.

Streaming software (OBS, Streamlabs, Discord) records exactly what is on your screen, so anything hidden for your viewers is hidden for you too. The **Fusion Recorder** is different: it records only the game picture, so your HUD, chat, phone and menus are never in its clips even while you can see them.

Video mode gives a clean screen for filming: no HUD, chat, minimap, name tags, prompts or notifications. *Video Mode Options* add letterbox bars (1.85:1 or 2.39:1), color filters (filmic, warm, cool, faded, high contrast, black & white ...) with a strength setting, and slow motion (50 %, 25 %, 10 %; only on your own screen). Also `/videomode`.

Free camera (`/freecam`): fly the camera with WASD and the mouse, Q / E down / up, Shift fast, Ctrl slow, scroll to zoom, Z / C to tilt, X to level it, Backspace to leave. Save **shot A** and **shot B** and **Play Camera Move** for a smooth glide between them (3 to 25 seconds); start recording first to film it. Players' cameras stay within *Free camera range* of their character (40 m by default) so it can't be used to spy; staff with `media.freecam.unlimited` have no limit.

Fusion Recorder: Start Recording (or `/record`; bind a key in the key bindings) records the game picture in 720p, 1080p, 1440p or full resolution at 30 or 60 fps, up to *Longest recording* (5 minutes by default). A small REC light shows while it records; it is not in the clip. Clips are uploaded to the server while you record and appear in **My Clips** (`/clips`):

- **Play** them, **Download** them to your PC (opens in your web browser and saves to Downloads), **Rename** and **Delete** them.
- **Edit:** trim the start and end, set slow motion (down to 0.25×) or fast forward (up to 4×), fade in / out and add a caption. **Add** more clips to join several into one. **Export as new clip** plays the edit once and saves the result as a new clip; your originals are kept.

Clips have no sound: the game page cannot hear the game. For clips with sound, use the **Rockstar Editor** in the same menu: **Record (Rockstar Editor)** starts and stops a Rockstar clip, **Action Replay** keeps the last few seconds so you can save something that just happened, and **Open Rockstar Editor** edits and exports them (Documents › Rockstar Games › GTA V › Videos). Opening the Rockstar Editor disconnects you from the server.

For the owner (Settings › Streaming & Video): turn each part on or off, the longest recording, the highest quality players can pick, storage per player (2 GB by default), how many days clips are kept (30), and the free-camera range. Clips are stored in `fusion-core/media-clips` on the server and are never part of an update or a release. **Install ffmpeg on the server** (Windows: `winget install ffmpeg`, Linux: `sudo apt install ffmpeg`) so clips are converted to MP4, which opens in every video editor (Premiere, DaVinci Resolve, CapCut) and on phones. Without it clips stay WebM, which plays in browsers, VLC and OBS but not every editor. Players download clips from the server's address on port 30120; if players join through a proxy or a domain, set *Clip address override*.

Permissions: `media.streamer`, `media.videomode`, `media.freecam`, `media.record`, `media.rockstar` (everyone by default) and `media.freecam.unlimited` (admins).

Reports and admin requests

- Players open **Menu › Help & Reports** to request an admin to their location, report a player, or send a question / bug report. Chat shortcuts: `/calladmin <reason>` and `/report <message>`.
- Staff with `admin.reports` get an alert with a sound. **F10 › Reports** (or `/reports`) lists open reports, with a red badge showing how many. Staff can **Claim** (the player is told who is helping), **Go to** the player or the report location, **Bring** them, **Message** them, open their player page, and **Close** with a note the player sees. Reports are kept in the database and in **Logs**.
- **Settings › Reports & Help:** turn each report type on or off, set the cooldown and the maximum open reports per player, and choose whether staff hear a sound.

Map blips (police stations, fire stations, anything else)

Settings › Map Blips comes with every vanilla police and fire station already on the map.

- **Add a location** (e.g. a new Paleto Bay police MLO): stand at it, click **Add**, give it a name and pick a **Type** (police, sheriff, fire, hospital, courthouse, city hall, mechanic, or custom). The icon appears on everyone's map **immediately**, with no restart.
- **Moved or removed a station?** Click **Here** to move it, uncheck **Shown** to hide it, or delete the entry.
- **Custom** blips take any icon and color number from the FiveM blip list.
- A blip can be limited to one job (e.g. a private police garage only police see).
- *Only show on the minimap when nearby* keeps the minimap tidy. Blips always show on the full map.

Banks, garages, stores, gas stations and hospitals put their own blips on the map from their own location lists, and those update live too.

Player blips (Settings › Gameplay › Players) cover the whole map, updated every 2 seconds:

- *Player blips on map*: staff with `admin.blips` see every player as `[id] name` (or set it to everyone, or off).
- *Emergency services see each other on the map*: on-duty members of the jobs in *Jobs that share emergency blips* (police, sheriff, ambulance and fire by default) see each other, blue for police, red for EMS, orange for fire, with a car icon when driving.
- Anyone can hide their own player blips with `/playerblips` (again to show them).

Loading screen

Settings › Loading Screen: title, subtitle, logo, accent color, layout (left / centered / minimal), background images (slideshow), background video, darkness, music playlist and volume, rotating tips, rules, staff list and links.

Media goes in `resources/[fusion]/fusion-loadscreen/html/media/`. After adding files, run `ensure fusion-loadscreen`, then type the file names (e.g. `city.jpg`, `theme.mp3`). URLs work too. Reconnect to see it.

Player guide

A short slideshow shown **once per player** (per Rockstar account) before they pick a character. The default pages cover the same ground as the Fusion Player Guide PDF: first join, controls, the menu, money and inventory, vehicles, phone and voice, homes, jobs, streaming and getting help.

- **Settings › Player Guide:** title, pages (title, text, picture), whether it can be skipped, and whether it appears in the main menu as *Server Guide*.
- **Formatting in page text:** `**bold**`, lines starting with `-` become bullet points, `[M]` shows a key cap, a blank line starts a new paragraph. `{server}` and `{player}` are replaced with the server and player name.
- **Changed the guide?** Raise **Guide version** by 1 and everyone sees it once more on their next join.
- Players reopen it any time with `/guide`.

Characters and spawning

- **Settings › Characters:** multiple characters on/off, number of slots (plus extra slots for `characters.extraslots`), deleting and switching characters, nationalities list, **starter items** for new characters, and whether new characters open the appearance editor.
- **Settings › Spawning:** spawn mode (**spawn selector**, last location, or always the default spawn), whether new characters use the selector, the default spawn point, *Last Location* in the selector, camera height, and the **selector locations**. Each location can be limited to one or more jobs (*Only for these jobs*) or a permission.
- **Settings › Teleport Locations:** the staff teleport list in *Menu › Misc Settings*.

Economy and banking

- **Settings › Economy:** the master switch, money types and starting amounts (cash, bank, crypto, or add your own), whether a type may go negative, paycheck interval and account, paying from society funds, bank and ATM locations, ATM prop models, transfers, and giving cash.
- Paychecks come from each job grade's pay (set in **Jobs**) and are paid on duty, or off duty if the job allows it.
- Bosses manage society funds at any bank (*deposit / withdraw*).

Jobs, gangs and boss menus

- **Jobs / Gangs tabs:** create jobs, add grades with pay, and mark boss grades. Job *type* (`leo`, `ems`, `mechanic`…) is what QBCore scripts look at.
- **Settings › Jobs & Gangs:** default job, remembering duty, toggling duty and opening the boss menu from the menu anywhere, the **job center** list (jobs players can take themselves), and **duty / boss locations**.

- Players use **Menu › Job**: duty toggle, job center, and the boss menu (hire the nearest player, promote, demote, fire, society balance).

Inventory and items

- **Settings › Inventory:** carry weight and slots, hotbar, giving and dropping items, drop despawn time, trunk and glovebox sizes, which jobs may search players, robbing downed players, and **stashes**. A stash can be shared or **personal** (every character gets their own), and limited to a job or permission.
- **Items tab:** create any item. To make it consumable, fill in *When used*: hunger/thirst/health/armor/stress amounts, *Consume* = 1 to use it up, and an animation (eat, drink, smoke, bandage, pills).
- **Item pictures:** they live in `resources/[fusion]/fusion-inventory/html/images/`.
  1. Create the item in **F10 › Items › New item**. Its *Internal name* (the small gray text on each tile) is what the picture is matched to, e.g. `energy_drink`.
  2. Put a PNG named after it in that folder: `energy_drink.png`. A transparent background around 100×100 pixels looks best.
  3. Type `ensure fusion-inventory` in the txAdmin console. New files are only picked up when the resource restarts.

  Every default item and weapon already comes with its own picture, made for Fusion Menu, so you only need to add pictures for items you create. Putting a file with the same name in the folder replaces the built-in one.

  Weapons share one picture per type (handguns, SMGs, rifles, shotguns, snipers, machine guns, heavy, melee, throwables, misc), saved as `weapon_<name>.png` so you can replace any single weapon's picture. Addon weapons added in game use `weapontype_<type>.png` until you give them their own. An item with no picture shows its category's picture (`category_<category>.png`), and if that is missing too, an icon.

  The picture then shows everywhere: inventory, item pop-ups, stores and the admin panel. To use a different file name, or a web link (no restart needed), open the item in **F10 › Items** and set **Image**. Names must match exactly, including lowercase: on a Linux VPS `Energy_Drink.png` and `energy_drink.png` are different files. Until an item has a picture it shows its category icon. Items added by QBCore scripts use the same folder (copy in the images that come with the script), and qb-inventory's pictures work as-is because Fusion uses the same item names.
- **ID cards and licenses:** using the *ID Card*, *Driver License*, *Weapon License* or *Pilot License* item shows the card itself to you and anyone within 3 m: the character's photo (taken from them as they are now), name, date of birth, sex, nationality and ID, while your character holds it up. **Settings › ID Cards & Licenses** changes each card live: the big name at the top (e.g. *State of San Andreas* → your city's name), the card title, its color, a seal or logo and a background picture (**Upload** or a link). Give licenses from the player's inventory in **F10 › Players**.
- **Weapons and ammo (inventory mode):** a weapon holds the ammo stored on it. Press **R** with an empty weapon to load a matching ammo item from your inventory and reload, or use the ammo item. To put a weapon away, press its hotbar number again (or use it again from the inventory).
- Players: **I** opens the inventory. Drag to move, right-click to use, drop items on *Use / Give / Drop*, set an amount to split a stack. At a vehicle's trunk it opens the trunk; inside a vehicle, the glovebox.

Vehicles, garages and keys

- **Spawner:** every vanilla and **addon** vehicle appears automatically, sorted by class. Addon packs (anything in `resources/[addons]` with a `vehicles.meta`) also get their own **Addon Vehicles** section. Packs started while the server is running (`ensure mypack`) show up within a few seconds, without players reconnecting. **Settings › Vehicles** controls spawning inside, replacing the previous vehicle, spawn cost (flat or catalog price), cooldown, plate format (`#` digit, `@` letter), list or picture-grid layout, saved vehicles, keys, locking NPC cars, the seatbelt, ejection speed and whether ejection hurts. Without a seatbelt a hard crash throws the driver or passenger through the windshield at the speed they were going, over the hood, ragdolling; with the seatbelt on nobody goes through. *Ejection injuries* decides how bad it is: **seriously injured** (the default: left barely alive, and fatal 60 km/h above the ejection speed), **always fatal**, or **light**.
- **Persistence (Settings › Vehicles › Persistence):** a vehicle a player spawned or drove stays in the world when everyone walks away. It is removed when it sits unused longer than *Remove after sitting unused* (5 minutes by default, 0 = never), when staff delete it, when its player leaves the server (can be turned off), or when the server restarts.
- **Taken off the server:** when a vehicle's pack is removed from the server, or the vehicle is deleted in **F10 › Vehicles**, it disappears from its owners' garages and impound the next time they log in or open one, and with the economy on its catalog price goes back to their bank.
- **Vehicles tab:** rename vehicles, set brand, price and class, hide a vehicle from the spawner, or restrict it to a permission or job (e.g. police cars for `police` only).
- **Vehicle Options** (in a vehicle): repair, wash, flip, god mode, engine, doors, windows, seats, **extras**, **liveries**, mod menu, wheels, colors, neon and headlights, plate text and style, torque/power, speed limiter, give keys, delete.
- **Settings › Garages & Impound:** garage and impound locations (car, air or boat; optional job), shared garages, **storing spawned vehicles** (they become owned, mods and all), max owned per character, impound fee, and sending vehicles left out to the impound on restart. **Add garage here** / **Add impound lot here** creates one where you stand, and **Here** moves a point (the interaction marker, or the spot cars come out, facing the way you face). Everything changes live for every player: markers, blips and menus, no restart. A garage or impound menu closes when you walk away from it.
- **Vehicle pictures:** `resources/[fusion]/fusion-vehicles/html/images/<model>.png`, then `ensure fusion-vehicles`.

Dealerships (Settings › Vehicle Dealerships): where players buy cars, aircraft and boats. Comes with Premium Deluxe Motorsport (cars), Elitas Travel at LSIA (planes and helicopters) and the Puerta del Sol marina (boats).

- **What is for sale comes from the vehicle catalog (F10 › Vehicles):** every vehicle with a price, sorted by class, picked by the dealership's type: *Cars, bikes and trucks*, *Aircraft*, *Boats* or *Everything*. *Only these classes* narrows it further (e.g. a motorcycle dealer, or a police dealership with the Emergency class and a job). New and add-on vehicles are for sale as soon as they are in the catalog; change a price, hide a vehicle or limit it to a job or permission there, and dealerships follow at once. Without a class list, car dealerships never sell service, emergency or military vehicles.

- **Buying:** walk up to the green marker, press the interact key, pick a class, and look through the vehicles. The one you are on stands at the dealership for you to see. Buying registers it to you (with its keys) and it comes out at the dealership; park it in any garage to keep it there. *Price multiplier* scales every price, and with the economy off vehicles are free (optional).
- **Adding or moving a dealership, live: Add dealership here** where customers should stand, then stand where bought vehicles should appear, facing the way they should point, and click **Here** next to *Vehicle comes out here*. Blips, markers and stock update for everyone straight away, no restart.

Siren controller and police radar

Siren controller (any vehicle in GTA's emergency class, plus models you add):

KEY	ACTION
Q	Emergency lights on / off
R	Siren on / off (turns the lights on too)
Y	Next siren tone
E (hold)	Air horn

Players can change all of these in **Settings › Key Bindings › FiveM**. Controller players set their own buttons there too: the *Fusion: ... (controller, set your button)* entries have no default.

> **With the Fusion Lighting add-on** (sold separately), Fusion Lighting's controller takes over the lights and siren: stages, patterns, buttons, the setup editor and **F10 › Settings › Emergency Lighting**. The radar, plate reader and AI pull-over below keep working. Turn ULC off while it runs, because both switch the same extras (Fusion Lighting imports each pack's `ulc.lua` instead). See its own guide.

With the lights or siren on, **AI traffic ahead pulls over to the right** and stops, then drives on once you have passed (Settings › Emergency Vehicles › Siren: on/off and how far ahead).

ULC (Ultimate Lighting Controls) is included in `[standalone]`. ULC only shows its controller for vehicles whose pack it has been told about; Fusion does that for you: every started pack with a `ulc.lua` (or `data/ulc.lua`) is listed in `resources/[standalone]/ulc/config.local.fusion.lua`, and ULC is restarted when the list changes. Your own ULC `config.lua` is never touched.

An on-screen **siren controller** (a control head with lit LIGHTS, SIREN and HORN buttons and a button for every tone) sits next to the speedometer. Press **F6** to use it with the mouse while you keep driving; F6 again gives the mouse back. Hide it from **Menu › Emergency** (or with `/hud`). Each vehicle only gets the tones of its service, like real US units:

SERVICE	TONES (Y CYCLES)	AIR HORN
Police / sheriff	Wail, Yelp, Priority, Wail 2, Yelp 2	Air horn
EMS / ambulance	Wail, Yelp, Hi-Lo, Phaser	Air horn
Fire	Mech Wail, Mech Yelp (mechanical fire siren), Wail, Yelp	Fire truck horn

The vanilla ambulance and lifeguard are EMS and the fire truck is fire; addon vehicles are sorted by name (*amb*, *ems*, *medic*, *rescue* → EMS; *fire*, *ladder*, *pumper*, *tanker*, *brush* → fire), and everything else in the emergency class uses police tones. Change any of this in **Settings › Emergency Vehicles › Siren** (EMS models, fire models, always-police models). **Your own siren sounds:** put an `.ogg` or `.mp3` recording in `resources/[fusion]/fusion-core/html/sounds`, run `ensure fusion-core`, and add a tone in **Settings › Emergency Vehicles › Siren tones** with that file name as its *Recording* (a web link works too). Every nearby player hears it from the vehicle, louder as it gets closer. Air horns can use recordings too, and *Recorded sirens are heard up to* / *Recorded siren volume* tune them. Server-side siren packs (audio banks) still work by entering their sound name, sound set and audio bank instead. Everyone nearby hears the same siren, synced through the server. The same controls are in **Menu › Emergency**.

Police radar and plate reader:

KEY	ACTION
F5	Radar power on / off
Numpad 8 / Numpad 5	Lock / unlock the **front** / **rear** speed
Numpad 9 / Numpad 6	Lock / unlock the **front** / **rear** plate
Numpad 0	Start / reset the **PIT timer**
F6	Radar remote (and the siren controller) with the mouse

It works like a dual-antenna moving traffic radar:

- **What it reads:** each antenna (front and rear) shows the **strongest** return in its beam in the TARGET window (big and close vehicles win, so a truck can hide a car behind it), the **FAST** window when something faster is in the beam, and a **LOCK** window. Only moving vehicles return a signal. Each antenna watches **OPP** (oncoming), **SAME** (same direction) or **BOTH**; by default the front watches oncoming traffic and the rear watches traffic coming up behind. Readings are the speed along the beam, so a car off to the side reads a little low (the cosine effect), and walls and buildings block the beam. Arrows show whether a target is approaching (▲) or moving away (▼).
- **Locking:** a lock holds the faster of TARGET and FAST in the LOCK window while the live windows keep tracking. Front and rear lock separately. The speed alarm (with optional auto-lock) flashes the front target and beeps.
- **XMIT / HOLD:** on hold the antennas stop transmitting: nothing reads, locked speeds stay. Lamps show XMIT/HOLD, moving or stationary patrol (MOV/STA) and the range.
- **Remote (F6):** POWER, XMIT/HOLD, TEST (the self test also runs at power-up: all segments light, then PASS), SELECT front/rear (the antenna MODE and audio work on), FRONT/REAR antenna on-off, MODE (OPP/SAME/BOTH), FAST, LOCK/REL for each antenna's speed and plate (F SPD, R SPD, F PLT, R PLT), AUTO lock, RANGE 1-5, AUDIO (Doppler tone whose pitch follows the target's speed, three volumes), DIM, PIT and the speed alarm.

- **Plate reader:** reads the nearest plate in its camera view, moving or parked, when it has a clear line of sight; the plate is drawn in its own style (blue on white, yellow on black, yellow on blue...). A **read log** under the windows lists the last plates with the game time, BOLO hits in red. When it sees a BOLO plate it locks, beeps and shows the reason.
- **Radar Display** (*Menu › Emergency › Radar Display*, each officer's own): layout (full, compact, front antenna only), target digit color (red, amber, green, blue), size, brightness (auto dims at night), see-through, unlit "888" segments, and showing the plate reader, read log and PIT strip. Hide the whole display from **Menu › Emergency** or with `/hud`, and move it with `/hud`.
- **Menu › Emergency** also has transmit/hold, each antenna's mode, range, fast window and audio, **Run plate** (registered owner and model from the database) and the **BOLO list**.

Keys besides the ones above (XMIT/HOLD, select antenna, lock/release the selected antenna, antenna mode, audio volume) start unbound, since lighting scripts often use the numpad: set them in *GTA Settings › Key Bindings › FiveM*.

PIT timer: when a pursuit starts, press **Numpad 0** (or *START PIT* on the remote, or *Menu › Emergency › Start PIT Timer*). A strip under the radar windows counts the pursuit time: amber *WAIT 0:32* until the department's time is up, then green **PIT AUTHORIZED** with a chime. If the department sets a top PIT speed it shows *SLOW TO 60 MPH* in red while you're going faster, and it shows *PIT NOT ALLOWED* when the department doesn't allow PITs at all. Press Numpad 0 again to stop and reset it.

Each department controls its own PIT rules. Its command staff (any boss grade, or the department's *command grade* and up, grade 3 by default) open *Menu › Emergency › Department PIT Rules* and set: whether PITs are allowed, how many seconds into a pursuit they're authorized, the top speed for a PIT (0 = no limit) and, for bosses only, which grade counts as command. Changes apply straight away, and everyone in that department gets a message with the new rules. Admins set every department in **Settings › Emergency Vehicles › PIT timer** (also the default time for departments without their own rule, and the switch for the whole feature). Staff with `emergency.pit` can change their own department's rules without command rank.

Settings › Emergency Vehicles: turn the siren controller and radar on or off; which jobs may use each (empty = anyone for sirens; police by default for the radar) and whether they must be on duty; extra models for unmarked cars and addon packs outside the emergency class; whether the front passenger can work the siren; what happens when the driver gets out; the **siren tones** (name, sound, and optional sound set / audio bank for server-side siren packs, limited to certain models if you like); the air horn sound; radar range and beam width; plate reader range; and the BOLO list.

Permissions: `emergency.siren` (everyone by default), `emergency.radar` and `emergency.bolo` (radar and BOLOs regardless of job; admins by default), `emergency.pit` (change your department's PIT rules without command rank; admins by default). Everyone can rebind the keys in *GTA Settings › Key Bindings › FiveM*, and move both panels with `/hud`.

Phone

Players open the phone with **F1** (rebindable) or by using the **phone** item from the inventory. Every character has their own number (the same one QBCore scripts see).

APP	WHAT IT DOES

Phone	Keypad, recent calls and contacts. Calling rings the other player; the call connects with voice through **Fusion Voice** (or pma-voice / mumble-voip if you use those instead)
Messages	Text conversations with other players, with unread counts and a new-message alert
Contacts	Add, edit and delete contacts; call or text them in one tap
911	Dial or text the emergency number to describe what's happening. On-duty police / EMS get the call with your name, number, **postal** and a flashing map marker
Bank	Bank and cash balance, and sending money to another player's number
Alerts	Every emergency alert received. Police (and staff) can issue an **AMBER Alert** from here
Weather	The current conditions and temperature, what is coming next (the server plans its weather ahead, so the forecast is real), and any severe weather warning
Web	A browser with the bookmarks you set (Discord, rules, wiki, store...). Optionally players can type any address. Some websites refuse to be shown inside apps and stay blank
Settings	Your number, wallpaper, and turning AMBER Alerts on or off (other emergency alerts cannot be turned off, like a real phone)

Emergency alerts appear as a banner at the top of the screen with the real two-tone alert sound (853 + 960 Hz), on every phone at once. Severe weather events send a warning automatically; an AMBER Alert lists the child, suspect, vehicle and plate, and the plate goes straight onto the police radar's **BOLO** list. Staff can send any alert from **F10 › Alerts**, which also shows the history.

Settings › Phone: turn the phone on or off, whether players need the phone item (and which item), which apps are available, wallpapers (built-in or your own pictures), how long calls ring, how long texts are kept, the emergency number, which jobs receive 911 calls and whether they must be on duty, how long 911 calls stay on the map, automatic weather alerts, the alert sound and volume, whether players without a phone still see alerts, which jobs may issue AMBER Alerts, and whether AMBER plates become BOLOs. Permissions: `phone.use`, `phone.amber`, `admin.alerts`.

Voice and radio (Fusion Voice)

Fusion has its own voice system built on FiveM's voice chat, so no pma-voice is needed. Players talk with their normal push-to-talk key (GTA Settings › Voice Chat).

- **Ranges: F11** cycles Whisper / Normal / Shout (a circle shows the range). The HUD shows your range, radio channel, call and a green glow while you talk.
- **Radio: Menu › Radio** or `/radio 12` to join a channel (`/radio 0` to turn it off), then hold **Caps Lock** to talk. Everyone on the channel hears you anywhere on the map, with a radio effect and mic clicks. Restricted channels (e.g. 1–10 for police and EMS) and needing a radio item are settings.
- **Calls:** phone calls are heard clearly from anywhere.
- QBCore scripts written for pma-voice (radios, phones, MDTs) work with Fusion Voice automatically.

Settings › Voice & Radio: turn Fusion Voice on or off, the ranges and distances, the starting range, the HUD, Fusion's radio on or off, the radio item, the highest channel, restricted channels and their jobs, radio and call volume, the radio effect, mic clicks and the radio animation. If pma-voice is running, Fusion Voice switches itself off.

Voice without Fusion's radio: turn off **Settings › Voice & Radio › Fusion radio**. Proximity voice and phone calls keep working; the Radio menu entry and `/radio` go away and everyone on a channel is taken off it. A radio script of your own that uses the pma-voice functions (qb-radio and the like) can still put players on channels, and Caps Lock talks on them.

Using pma-voice instead of Fusion Voice:

1. Download pma-voice from https://github.com/AvarianKnight/pma-voice (**Code › Download ZIP**), unzip it into `resources/[standalone]` and rename the folder from `pma-voice-main` to `pma-voice`.
2. In `server.cfg`, above `ensure [qb]`, add pma-voice's settings and start it:

```
setr voice_useNativeAudio true
setr voice_useSendingRangeOnly true
setr voice_defaultCycle "GRAVE"
ensure pma-voice
```

3. **F10 › Settings › Modules**: turn **Voice and radio** off (or just leave it: Fusion Voice turns itself off while pma-voice runs).
4. Restart the server.

Phone calls switch to pma-voice by themselves, and scripts written for pma-voice talk to the real pma-voice. Fusion's own radio (`/radio`, the radio in the M menu, restricted police channels) and the voice indicator on Fusion's HUD belong to Fusion Voice, so with pma-voice use a pma-voice radio script (e.g. qb-radio in `[qb]`) instead. To go back, remove pma-voice and its `server.cfg` lines and turn **Voice and radio** on again.

Speed limits

Every road has a speed limit. Drivers see it on a **sign next to the minimap** (US "SPEED LIMIT" or a European red circle; players move or hide it with `/hud`). The sign turns red while they're going faster.

Settings › Speed Limits: turn it on or off, the sign and its style, and the limits themselves:

- **Default for each kind of road**, from the road's name: freeways (65 mph), highways and routes (55), avenues and boulevards (40), streets in the city (35) and roads in Blaine County (45).
- **Roads with their own limit**: a starter list of GTA's freeways, highways and main roads (Route 68, Great Ocean Hwy, Senora Fwy ...). Change any of them, delete them, or add more by the name the HUD shows. They win over the defaults.
- In game, staff with `admin.speedlimits` stand on a road and type `/speedlimit 85` to set its limit, or `/speedlimit reset` for its default. Anyone can type `/speedlimit` to see the road's limit.
- *Limits below are in* sets MPH or KM/H; the sign and radar show them in the server's speed unit.

Changes reach every player at once, and the **police radar** reads them too: with its speed alarm on *Over the posted limit*, an officer set to +20 is only alerted when a target goes 20 over the limit **of the road that target is**

on. Raise Route 68 to 85 and the alarm follows straight away. The radar shows the limit where the patrol car is (*LIMIT 65*).

Postal codes

The nearest **postal code** shows next to the street name on the vehicle HUD and on the on-foot location line (*Menu › Misc › Show Location*), and on reports and 911 calls. Type `/postal 1234` to set a waypoint to a postal, or `/postal` to clear it.

Settings › Postal Codes: turn it on or off, choose the postal set (*New postals*, *OCRP postals*, *Old postals*, or your own file in `fusion-core/data/postals`), show the distance, the `/postal` command, and postals on reports. **Fusion's postal map** (`fusion-postalmap`) draws the same postal set on the pause-menu map, so the map always matches the HUD and `/postal`. Turn off *Draw the postal codes on the pause map* if you prefer another postal map pack in `resources/[addons]`. For your own postal file, make its map with `tools/postal-map` (see the README there). Scripts written for *nearest-postal* (CAD/MDT systems and others) get their postal from Fusion automatically.

Postal map (F10 › Postal Map)

The pause map is Fusion's own illustrated map, with the postal codes and place names drawn on by the server. In **F10 › Postal Map** you can drag postals and names around, add a postal by clicking the map or **where you stand**, rename or delete them, and start from the New, OCRP or Old set. **Save & redraw map** redraws it (a few seconds) and reloads it for every player; the HUD, `/postal`, reports and 911 calls use the same postals. Players get the map as six streamed pause-map tiles (`fusion-postalmap/stream/minimap_sea_*.ytd`, about 2 MB each, made by the server from your map, so there is nothing to install). Players already in the game see a redraw after they reconnect. *Draw the postal codes on the pause map* (**Settings › Postals**) off brings back GTA's own pause map. In game, `postalmapcheck` in the F8 console shows whether a player's game has Fusion's tiles (2048×2048) or GTA's own (1024×1024). **Settings › Postal Map** turns the place names on or off and sets the postal badge size and color.

Housing (F10 › Housing)

Apartments and houses players buy and live in, with real GTA interiors they walk around in. Every apartment and house is its own private copy of its interior, so a hundred homes can use the same style without anyone meeting.

Two kinds of location:

- **Apartment building:** one entrance, as many apartments as you allow (*Apartments in this building*, 0 = no limit). Each player buys their own apartment there; each one has its own owner, keys, lock, doorbell, storage and garage.
- **House:** one owner.

Placing a home (staff, `admin.housing`):

1. Stand **outside at the real front door** (the building's lobby door, the house's front door) and press **F10 › Housing › Add a home here**: name, apartment building or house, how many apartments, interior style and price. Fusion refuses to place an entrance inside a home, a preview or an interior.

2. **Garage:** sit in a car where vehicles should come out, facing the way they should drive off, and press **Set garage here** on that home. Homes without one show *No garage yet*. Staff see every home's garage spot as a purple car marker; players see it once they own the home or hold a key.
3. **Ways out inside (Exits):** press **Go inside** on the home, then open **F10 › Housing › Exits** on it (or the **Exits** button in the *Inside* bar at the top). Stand just inside the front door, facing it, and press **Move it here** (players arrive there too). Add more exits for balconies or back doors with **Add an exit where I stand**, and **Remove** them again. Every home location has its own exits, like its own entrances. Until a home's front door is set, it uses the interior style's default (set with **Preview style › Move the front door here**); **Reset** goes back to that default.

More ways in: a building with a front door and a garage around the back gets an entrance at each. Press **Entrances** on the home, stand outside at the other door and press **Add an entrance where I stand** (give it a name like *Garage side*). Every entrance works the same, and players come back out where they went in. A **balcony** can be an entrance too: walk out onto it (also from inside the house) and add it there. Fusion only refuses spots the game says are indoors. Residents then pick *Leave by the balcony* at an exit to step out onto it. Many GTA balconies (and some roofs) have no floor you can stand on. Where an entrance has nothing under it, Fusion puts an invisible floor of about 3.7 x 3.1 m there, level with the marker, so nobody falls through. Stand where the floor should be when you add the entrance (in noclip, hover at standing height above the balcony).

Each home also has **Edit**, **Move entrance here**, **Entrances**, **Exits**, **Set / Move garage here**, **Go to**, **Go inside** (that home's own interior, to set its exits), **Residents** (apartment buildings: who lives there, **Evict** one apartment), **Evict** (houses) and **Delete**.

Interior styles (Settings › Housing › Interior styles): a studio and a mid-range apartment, five high-rise apartments (4 Integrity Way, Del Perro Heights, Richards Majestic, Tinsel Towers, the Eclipse Towers penthouse) and eight Vinewood Hills houses come ready. Add your own (any GTA interior or an MLO you've installed) and set its front door the same way.

For players:

- A green marker is something for sale: **Look inside**, then **Buy** (or **Claim** when the economy is off). A character can own up to *Homes one character can own* (default 2), and one apartment per building.
- At an entrance: **Go inside** and **Lock / Unlock** your own home; homes you hold a key to; and in apartment buildings **Residents**, to ring someone's doorbell (or walk into an unlocked apartment). Whoever is inside sees **Let in** in **M › My Home**.
- **Garage:** at the home's garage spot, **park** your vehicle (in it) or **take one out** (on foot). Owners and key holders can use it; everyone only sees their own vehicles there. *My Vehicles* shows them as *Garage: (home name)*.
- **Storage and wardrobe spots:** the owner walks to a shelf, cupboard or closet and picks **Put my storage here** / **Put my wardrobe here** in **M › My Home**. A marker shows the spot and **E** opens it: storage is the home's stash (slots and weight from the interior style), the wardrobe is the clothing editor with saved outfits, free. Key holders use them too. Staff can give a whole interior style default spots: preview it (F10 › Housing), stand at the spot and pick **Set this style's storage / wardrobe here** in **M › My Home** (or type them into the style in **Settings › Housing › Interior styles**). An owner's own spot wins over the style's, and **back to the default spot** undoes it. Homes without any spot keep **Storage** and **Wardrobe** in **M › My Home**.

- **M › My Home**: inside, **Storage** and **Wardrobe** (when the home has no spot for them), visitors to let in and **Leave**; for each home, waypoints to it and its garage, **Give a key**, **Key holders** (and **Take back**) and **Sell** (refunds *Selling back refunds* % of the price).
- **Spawn at home:** owned homes are listed on the spawn screen; picking one starts you inside.
- Police (*Jobs that can force entry*, on duty) get **Force entry** at locked doors; the owner is told.

Leaving: at any exit, players choose where they come out: any of the home's entrances, or straight to its **garage**. With only one choice they just leave. Logging off inside brings you back outside the entrance. Selling, eviction and deletion put everyone inside out, every key stops working, and vehicles in that home's garage move to the first public garage. Storage belongs to the owner, so a new owner starts with an empty one.

Vinewood sign (F10 › Vinewood Sign)

The sign on the hill is Fusion's own: two rows of up to **12 characters** (letters, numbers and `! ? . - ' &`), the back row higher up the hill behind the front row. Pick a color per row (or a fade from one color to another), the night effect (steady glow, slow pulse, chasing lights, sparkle, rainbow or none), whether it lights up the hillside at night, the hand-built look (slightly uneven letters like the real sign) and the daytime brightness. The editor shows a live day and night preview; **Save to the sign** changes it for everyone straight away. The letters stand on legs that run into the hill, so nothing floats, and GTA's own sign is gone at every distance while the Vinewood module runs (turn the module off to get GTA's VINEWOOD back).

Settings › Vinewood Sign › Letters picks how they're made:

- **3D models** (default): solid letters in a real sign typeface (Oswald Bold), lit by the sun by day and glowing by themselves at night, with every color, fade and night effect.
- **Drawn letters**: simple flat-shaded letters, with adjustable letter sizes.

Colors: click a color box to open the color picker (any color, plus quick swatches). The same picker is used for every color in F10.

Slurs and hate words are always refused (also when spelled with numbers or symbols, spaced out or split over both rows), and online staff get a warning naming who tried. **Settings › Vinewood Sign** sets letter sizes and positions, also blocking swear words, your own blocked and always-allowed words, and the staff warning. Permission: `admin.vinewood`.

Handling editor

F10 › Handling edits an addon vehicle's handling and saves it into the pack's own `handling.meta` on the server.

1. Get in the vehicle and click **Load my vehicle** (or type its model name). Fusion finds its handling entry through the pack's `vehicles.meta`, even when the handling has a different name than the model.
2. Change any line: mass, drag, center of mass, gears, power, top speed, brakes, steering lock, grip, suspension, anti-roll bars, damage multipliers, flags and more. Each line shows what it does and its original value. Use **Filter fields** to find one quickly. Each line also shows its **safe range**; values outside it turn red and are refused, because they break the vehicle in game (for example a center of mass far below the car makes it hover).

3. Click **Save to server files**. Only the lines you changed are rewritten, and the original file is backed up first.
   - With **Apply now** checked (the default), the pack is reloaded straight away and you are respawned in the vehicle with the new handling. Vehicles of that model on the server are removed first, because reloading a pack while its vehicles exist can crash players.
   - Without it, the change applies at the next restart.
4. **Backups / undo** restores any earlier save with one click.

Vanilla GTA vehicles keep their handling inside the game files, so only addon vehicles can be edited. Protected (escrowed) packs are shown read-only. Needs the `admin.handling` permission (admins have it). Like the carcols fixer, a newly added pack can be edited after the next server restart.

Vehicle meta (carcols) fixer

> **With the Fusion Lighting add-on**, its Lighting Fixer does this job by default, and this page's Scan and Fix are turned off (Backups / undo still work). **F10 › Settings › Emergency Lighting** › *Fix pack ID conflicts with Fusion Lighting instead of Fusion's Sirens & Carcols page* switches back.

When two vehicle packs use the same **siren**, **light settings** or **modkit** ID, cars get the wrong sirens or lightbar patterns, or lose liveries and mods. **F10 › Sirens & Carcols** (its own page in the admin panel) solves this:

1. **Scan vehicle packs:** reads every resource's `carcols.meta` and `carvariations.meta` and lists each clashing ID, which pack keeps it, and which packs need a new one.
2. **Fix:** gives the clashing packs unused IDs and updates their `carcols.meta` and `carvariations.meta` together, so vehicles keep their own sirens, lights and mods. IDs keep their style: small siren/light IDs stay within 100-255 for older game builds, large ones stay large.
3. **Restart the server** so players download the fixed packs.

- Every changed file is backed up first (`fusion-core/carcols-backups`). **Backups / undo** restores any fix with one click.
- **Protected (escrowed)** packs can't be edited. The tool keeps their IDs and changes the other pack instead; a clash between two protected packs is reported for their authors to fix.
- Needs the `admin.carcols` permission (admins have it). FiveM only lets Fusion edit a pack that is listed in `resources/[fusion]/fusion-core/permissions/filesystem.cfg` ; Fusion rebuilds that file at every start and `server.cfg` runs it. A pack you add is editable **after the next restart**.

Weapons

- **Settings › Weapons:** free mode (freeroam style) or inventory items, ammo given, weapons hidden from the spawner, keeping weapons after death and relog, replacing GTA's TAB wheel with the Fusion wheel, **job loadouts** (pick weapons, jobs, optional permission, on-duty only, auto-attachments), and **gas grenades**: BZ gas and tear gas impair every player in the cloud (blurry, shaking screen, coughing and eye-wiping, no sprinting) until a few seconds after they get out. Set the cloud radius and how long it lasts, whether players in vehicles are affected, and which mask numbers count as gas masks.

- Players get **Menu › Weapons**: spawner by category, attachments, tints, ammo, unlimited ammo / no reload (staff), personal loadouts with *equip on spawn*, and the weapon wheel (`/wheel`).
- Weapon categories are permissions (`weapons.category.heavy` ...), so you can, for example, allow handguns to everyone and rifles only to police.
- **Addon weapons:**
  1. Put the weapon pack in `resources/[addons]` and restart.
  2. Add it under **Settings › Weapons › Custom / addon weapons** with its spawn name (e.g. `weapon_glock19`), display name, category and ammo type.
  3. It appears in the spawner, becomes an inventory item, and is included in `QBCore.Shared.Weapons` . Add a picture as `fusion-weapons/html/images/<spawn name>.png` .

Appearance and EUP

- **Settings › Appearance & EUP:** editor anywhere vs stores only, clothing / barber / surgeon prices, **store locations** (a store with a job becomes a free locker room), **uniform presets**, and the ped model list.
- **Creating EUP uniforms:**
  1. Dress your character (Menu › Appearance › Clothing; **Set By Number** jumps straight to an EUP drawable). At the bottom of every editor menu, **Camera** keeps the camera on one part of the body while you go through the items (head, chest, hands, waist/belt, legs, feet, back, full body; *Auto* follows what you highlight), which helps with pieces in an unexpected slot such as a holster on the bag/parachute slot. **Turn Character** turns the character 45° with ← → (ENTER turns it around). Holding A and D turns it freely (FiveM key bindings "Fusion appearance: turn the character left / right").
  2. Go to **Appearance › Uniforms › Save Current Outfit As Uniform**, and give it a name, department, jobs and a minimum grade.
  3. Fine-tune or add a picture later in **Settings › Appearance & EUP › Uniform presets**.
  4. After an EUP update (or to change a piece), put the uniform on, fix what changed, and pick **Update With What I'm Wearing** on that uniform. It keeps its name, jobs, grades and picture.
- **My Outfits** (Appearance › My Outfits): players save outfits, then **Wear**, **Update With What I'm Wearing** (re-save it after changing a piece or an EUP update), **Rename** or **Delete** them. Outfits made for another character model still open for renaming, updating and deleting.
- **Uniform pictures:** `resources/[fusion]/fusion-appearance/html/images/` , then `ensure fusion-appearance` .
- **Adding EUP or other clothing packs:** put the pack in `resources/[addons]` and restart. Its clothing shows up in the editor automatically, after the vanilla items in each slot. Use **Set By Number** to jump straight to a drawable.

Stores

Settings › Stores: catalogs (what is sold, prices, and an optional license such as `weapon`) and **store locations** (what it sells, blip, optional shopkeeper ped, optional job). Choose between a menu list and the inventory shop screen (drag to buy). Paying by card when cash is short is optional.

- Comes with 24/7 and LTD supermarkets, Rob's Liquor, You Tool hardware stores, Ammu-Nation, and Digital Den electronics, plus catalogs for each and a **Pharmacy** catalog (bandages, painkillers, first aid kits, IFAKs) ready for any store you add.
- **Adding a store, live:** stand at the counter, click **Add store here**, name it and pick what it *Sells*. It appears for everyone straight away with its blip, prompt and shopkeeper. **Here** moves a store; delete the entry to remove it. No restart.
- **A new kind of store:** add a catalog (e.g. *Bait Shop*) with its products, reopen F10, then add store locations that sell it.

Fuel

Settings › Fuel & Gas Stations: consumption overall and per vehicle class, price per liter, jerry can price, refuel speed, pump and EV-charger props, electric vehicles, low-fuel warning, NPC vehicle fuel, and station blips. Pumps work at every gas station automatically.

Running out: below *Engine sputters below* (5% by default) the engine stumbles, more often as the tank empties, then dies at 0 and the car coasts. Electric vehicles lose power smoothly instead. Set it to 0 to switch sputtering off.

Jerry cans: bought full at a pump (G): the can's price plus the fuel in it. With fusion-inventory running a jerry can is an inventory item in either weapon mode, and it keeps the fuel left in it, so it can go in pockets, trunks and stashes. With the can in hand: **E** at a vehicle pours it in, **E** at a pump fills it (paying for the fuel it takes), and **G** puts it away with its fuel. *A full jerry can fills this much of a tank* (25% by default, about a real 5-gallon can) sets how far one can goes.

- **Refuel points (aircraft, boats and your own stations):** areas where the driver refuels **from the seat** with the engine off: stop inside one and press the interact key. Each has a type (*Aircraft*, *Boats*, *Cars*, or *Every vehicle*), a radius, an optional job and a blip (blue for aviation fuel, teal for marine fuel). It comes with aviation fuel at LSIA, Sandy Shores Airfield, McKenzie Field and the Pillbox Hill, Mission Row and Vespucci helipads, and marine fuel at the LSYMC marina.
- **Adding or moving them, live: Add refuel point here** where you stand, **Here** to move one, and the same for gas station blips (**Add blip here**). Everyone sees the change straight away, no restart.

Hospital and death

Settings › Hospital & Death:

- **Respawn mode (freeroam):** short timer, then respawn at the nearest hospital or where you died.
- **Downed mode (RP):** players go down, then die after a timer. EMS revive them with the revive item (first aid kit), or they respawn after bleeding out, paying a bill and optionally losing items (with a keep list). While down they can **call EMS** (G: every EMS on duty gets an alert with their location, once a minute; with nobody from the EMS jobs on duty, an **AI medic** drives over in an ambulance and revives them for the *AI medic bill*, 250 by default; turn it off or change the vehicle in **Settings › Hospital › EMS**) or **give up** (hold H for 3 seconds: they die and may respawn after *After giving up, respawn allowed after*, 30 seconds by default; turn giving up off in **Settings › Hospital › Downed mode**). Both are FiveM key bindings.

- **Fire** (Settings › Gameplay › Fire): walking into any fire (a molotov, a burning vehicle, Fusion Fires) or sitting in a burning vehicle for 2 seconds sets the player alight. They lose 12 health a second (adjustable) until the fire goes out, 5 seconds after catching fire or at once in water. Fusion god mode, noclip and spectate are spared.
- **Falling:** a fall longer than *Fatal fall height* (15 m by default) without an open parachute kills, even for staff (only Fusion's own god mode, noclip and spectating spare you); landing in water is survivable. High up (*Free fall starts this high above the ground*, 30 m by default), a falling player drops belly-down like a skydiver instead of flailing; without a parachute of their own they can't open one, and the landing is still fatal.
- **Pillbox Hill hospital you can walk into:** GTA has its own walk-in Pillbox lobby that multiplayer normally leaves switched off (only a painted fake interior shows behind the glass). *Open GTA's own Pillbox Hill hospital interior* (on by default, **Settings › Hospital › Locations**) switches it on. It is the only Pillbox interior in GTA, and the game only has it as a story mission left it, after a truck crashed through. Fusion streams a repaired copy (`fusion-hospital/stream/v_int_40.ytyp`): the debris, rubble, litter, hanging pipes and scorch marks are gone, the rusty waiting seats, bent bench seats and tipped-over armchairs are replaced with clean ones (and the rusty plastic chairs are gone), knocked-over furniture stands upright again, a scorched door is swapped for a clean one, and clean panels in GTA's own hospital finishes cover the damage built into the building (the torn-open corridor ceiling, the broken doorways, wall ends and sunroom window posts), the sunroom's broken window panes have glass again, and the light fittings left hanging at the ward hall are gone. The front entrance gets GTA's hospital glass doors, which swing open as you walk through. It also leaves out the painted fake interior and its closed doors (whose collision blocked the entrance). The default check-in desk and respawn point are inside. If you install a Pillbox MLO, turn this off and delete `fusion-hospital/stream/v_int_40.ytyp`, `fusion_pillbox.ytyp`, `fusion_pillbox.ytd`, `v_40_lights2.ydr`, `v_hospital.ybn` and the `fusion_pb_*.ydr` files.
- **Places that need an MLO:** a teleport or respawn to a spot with no floor outside every GTA interior puts you on the nearest sidewalk instead of dropping you through the map.
- EMS jobs, alerts to on-duty EMS, check-in desks (cost, treatment time, disabled when enough EMS are on duty), and hospital locations.
- **Treatment beds:** a checked-in patient is taken to a free bed, lies on it while being treated, then gets up beside it. Each hospital has its own *Treatment beds* list (**Add bed here** while lying on or standing on a bed, facing the foot end); Pillbox comes with its 8 ward beds. A hospital without beds treats patients at the desk.

Jail

Settings › Jail: jobs allowed to jail, the longest sentence, confiscating items (returned on release, with a keep list), time counting offline, cell, release point, escape radius, and what happens on escape. Police use **Menu › Jail** or `/jail <id> <minutes> <charges>`.

Door locks

Lock any door in the game: a police station, a business, a house, a gate.

- **Adding doors:** staff open **Menu › Door Locks**, look at a door and press *Add the door you are looking at*, then give it a name. For double doors, *Add a double door* asks for each half in turn. A new door starts locked.

- **Who may use it** is set per door in **Settings › Door Locks › Doors**: any number of jobs (only while on duty, unless you turn that off), a key item that opens it, or both. Staff with `doorlock.all` open every door; `doorlock.manage` (admins by default) adds and removes them.
- **Each door** also has *Locked after a server restart*, *Lock again by itself after* (seconds, for doors that should never stay open) and its reach. The position and model are filled in when you add it; you can type them too (a model name or hash).
- **Using a door:** near a door you may use, the prompt shows its name and whether it's locked; the interact key (E, a FiveM key binding) locks or unlocks it with a key-card swipe. Everyone sees the change at once, and a locked door can't be opened by anyone.
- **Other scripts:** `exports['fusion-doorlock']:SetDoorState(nameOrId, locked)` and `GetDoorState(nameOrId)`; the server event `fusion:doorlock:changed (id, locked)` fires on every change.

Weather and time

Settings › Weather & Time: dynamic weather and cycle length, allowed weather types, weather after restart, snow on the ground, blackout, day length, freezing time, and real-world time with an offset. Staff also get **Menu › Weather & Time** for instant changes.

Road conditions. Weather changes how vehicles handle: rain makes the roads wet (grip and braking drop, and roads take a few minutes to dry once it stops), snow cuts grip further, a blizzard leaves the roads icy, and driving fast through standing or flood water makes a car aquaplane. Off-road vehicles and SUVs cope better in snow. The HUD shows *Wet roads*, *Snow on roads* or *Icy roads* while you drive. Adjust or turn it off in **Settings › Weather & Time › Road conditions**: grip on wet, snowy and icy roads, drying time, the off-road bonus, aquaplaning and the HUD label. Only grip and braking change, scaled from each vehicle's own handling, and they return to normal when conditions clear or you get out.

Outdoor warning sirens. Civil-defense sirens on poles around the map sound for tornadoes and hurricanes (choose which events in settings). You hear the nearest one, from its direction, quieter indoors. Fusion makes the siren sound itself (*wail*, *steady*, *hi-lo* or *whoop*), or plays a recording you add to `fusion-core/html/sounds` (or a web link). **Settings › Weather & Time › Outdoor sirens:** the events, the signal for tornadoes and for other events, recordings, how long they sound and how often they repeat, volume, the siren poles (name, position, range) and showing them on the map.

Severe weather events. Staff start them from **Menu › Weather & Time › Severe Weather** (or `/weatherevent <type>` / `/weatherevent stop`), or let them happen at random. Everyone gets an alert banner with a countdown first, the HUD shows the time left, and every player sees the same event:

EVENT	WHAT HAPPENS
Thunderstorm	Heavy rain, strong wind, rough sea and lightning strikes around players: a visible bolt every few seconds, landing near each player in turn (they can hurt people on foot and stall cars)
Flood	Torrential rain while the sea, rivers and lakes rise (3 m by default): the docks, low coastal roads and riverbanks go under, then the water drains away

Hurricane	Winds build to maximum, huge waves, a storm surge, a power cut across the city, lightning, and gusts that shove vehicles sideways
Tornado	A spinning funnel travels across the map toward players, shown on the map with a danger zone. It pulls in and lifts vehicles, people and loose objects, and hurts anyone caught inside. Comes with hail and lightning
Blizzard	Heavy snow, snow covering the ground and roads, strong wind and low visibility, with optional icy roads
Hailstorm	Hail bouncing off the ground, denting vehicles left outside and hurting people caught in the open, with strong wind and lightning

Settings › Weather & Time › Severe weather: turn events on or off, the warning time, random events (chance per hour, which types, minimum players), each event's length, lightning strikes and damage, hail and hail damage, flood height and how fast it rises, hurricane surge, gusts and power cut, tornado size, speed, strength and damage, and icy roads. Permission: `weather.events` (admins by default).

> The tornado funnel is built from GTA's own smoke effects. If you prefer another look, change the particle asset and effect under *Severe weather* (advanced).

Targeting (third eye)

Hold **Left Alt** and look at something. Options appear, and clicking selects one. QBCore scripts add their options automatically. **Settings › Targeting** sets ray length, default distance, outlines and a zone debug view.

Anticheat

Fusion's anticheat runs mostly on the server, reading what the game reports about each player, so a cheat menu can't hide from it by blocking a script on the player's PC. **Settings › Anticheat** controls everything, and **F10 › Anticheat** lists what it has caught.

What it detects

DETECTION	DEFAULT ACTION	HOW
God mode	Kick	Player is invincible without `player.godmode` (downed/dead players are ignored)
Health / armor above maximum	Kick	More health or armor than the game allows
Invisible	Alert	Invisible without `player.invisible`, noclip or spectate permission
Teleport / speed hack on foot	Log	Moved further than possible. Logs only by default, because some QBCore housing and apartment scripts teleport players
Impossible vehicle speed	Alert	Cars and bikes faster than the limit (aircraft and boats are ignored)
Blacklisted weapon	Alert	Holding a blacklisted weapon without the spawn permission for its category. The weapon is removed
Entity spam	Kick	Spawning vehicles, peds or props faster than the limit. Extra spawns are blocked

Blacklisted model	Alert	Spawning a blacklisted vehicle, ped or prop. The spawn is blocked. Vehicles taken from the spawner or a garage are allowed
Explosion spam	Kick	More explosions than the limit. Extra explosions are blocked
Modded explosion	Ban	Invisible or boosted explosions, or blacklisted types (orbital cannon, railgun…)
Honeypot event	Ban	Triggered a fake event that only cheat menus use
Injected resource	Ban	A resource started on the player's PC that doesn't exist on the server
Anticheat client stopped	Kick	The anticheat stopped answering on the player's PC, or they stopped a Fusion resource
Cheat menu command	Ban	A known cheat menu's command exists on the player's PC
Weapon damage modifier	Kick	Weapon or melee damage boosted
Spectating without permission	Kick	Spectator mode without `admin.spectate`
Night / thermal vision	Alert	Without `misc.nightvision` / `misc.thermal`

Actions: *Off*, *Log only* (saved in F10 › Anticheat and Logs), *Log + alert staff* (staff with `admin.anticheat` get a notification with a sound), *Kick* or *Ban*. Bans go into the normal **Bans** list; set the length with *Ban length in hours* (0 = permanent).

Settings › Anticheat also has:

- **Thresholds:** god mode checks before acting, teleport speed, max vehicle speed, entity spawn limits and explosion limit.
- **Blacklists:** weapons, models, explosion types, honeypot event names and cheat-menu commands. Thresholds and blacklists are private and never sent to players' PCs.

Avoiding false positives

- Admins have `anticheat.bypass` and are never flagged. Give it to anyone else you trust (testers, developers) in **Roles & Permissions**.
- Start new detections on *Log only* or *Alert*, check **F10 › Anticheat** for a few days, then switch to kick or ban.
- If a QBCore script of yours uses one of the honeypot event names, remove that name from the list.
- If a job or script hands out a blacklisted weapon (for example a SWAT armory giving out grenade launchers), give that job's role the `weapons.category.<category>` permission or remove the weapon from the blacklist.
- Delete a false positive with the bin icon on its row (`anticheat.manage`).

> No anticheat catches everything. This one stops the common menus and abuse; pair it with active staff and the Reports system.

Logging and Discord

Settings › Logging: turn logging on or off, mirror to the console, retention days, skip categories, and **Discord webhooks** (one for everything, or one per category). Webhook URLs are private and never sent to players.

8. Adding content

The easy way (recommended): stop the server, then drag the script's or pack's `.zip` (or folder) onto `install-script.bat` (Windows) or run `bash install-script.sh /path/to/pack.zip` (Linux). It works out where it belongs (scripts go to `[qb]`; vehicles, EUP, maps and peds to `[addons]`), refuses scripts Fusion already replaces, and upgrades a script you already have **without losing your config changes** (section 11).

By hand:

YOU WANT TO ADD…	PUT IT IN	THEN
A QBCore script	`resources/[qb]/<script>`	Restart the server (or `ensure <script>`)
Addon vehicles	`resources/[addons]/<pack>`	Restart. They appear in the spawner and **Vehicles** tab by themselves
EUP / clothing packs	`resources/[addons]/<pack>`	Restart, then build uniforms in game
Maps / MLOs / peds	`resources/[addons]/<pack>`	Restart
Item / vehicle / weapon / uniform pictures	the module's `html/images` folder	`ensure` that module
Loading screen media	`fusion-loadscreen/html/media`	`ensure fusion-loadscreen`

New files inside a resource are only picked up when that resource restarts. That's why you run `ensure`.

On a Linux VPS: files you upload with WinSCP or FileZilla logged in as `root`, or unzip with `sudo`, belong to root instead of the user the server runs as, and Fusion can't save into them (carcols fixer, handling editor, postal map). The Linux installer handles this for you: every minute it gives anything in the server folder back to the server user, so you never need to fix ownership yourself. To avoid it entirely, log WinSCP in as the server user (e.g. `ubuntu`) instead of `root`. Installed before this was added? Run `sudo chown -R ubuntu:ubuntu /home/fivem/fusion` after adding content (with your own user and folder).

Scripts that must start in a certain order

`server.cfg` starts each of `[qb]` and `[addons]` with one line (`ensure [qb]`), and FiveM doesn't promise any order **inside** a folder. On Linux it often isn't even alphabetical. Most scripts don't care. For scripts that need another one running first (their readme usually says so):

1. Put them in `resources/[standalone]`, **not** in `[qb]` or `[addons]`. `ensure [qb]` would start, or restart, everything in `[qb]` in its own order.
2. Start them by name in `server.cfg`, in the right order, **above** `ensure [qb]`:

```
ensure screenshot-basic
ensure eyewitness
ensure overwatch
```

```
ensure [qb]
ensure [addons]
```

If a script's `fxmanifest.lua` already lists the other one as a `dependency`, FiveM handles the order itself and it can stay in `[qb]`.

screenshot-basic is included

Many scripts take screenshots of the game view: CCTV and evidence scripts (e.g. Eyewitness), MDTs, phone cameras. They need **screenshot-basic**. Fusion's server kit includes it, already built, in `resources/[standalone]/screenshot-basic`, and `server.cfg` starts it before `[qb]` and `[addons]`, so it's always running first.

- **Don't install a second copy** in `[qb]` or `[addons]`. Two resources with the same name conflict. If a script's instructions tell you to install screenshot-basic, it's already done.
- **Servers set up before it was included:** copy `resources/[standalone]/screenshot-basic` from the latest server kit into your server, and add `ensure screenshot-basic` just above `ensure [qb]` in `server.cfg`.

Organizing [qb] and [addons] into sub-folders

You can group content into sub-folders, as deep as you like, as long as **every grouping folder has square brackets**:

```
resources/[addons]/
├ [cars]/
│  ├ [police]/lspd-pack/      ✓ starts
│  └ [civilian]/supercars/    ✓ starts
├ [eup]/my-eup-pack/          ✓ starts
├ cars/mypack/                ✗ "cars" has no brackets: the server ignores everything inside
└ mypack/mypack/              ✗ zip extracted one folder too deep: move the inner folder up
```

A folder **with** brackets is a category (just for organizing). A folder **without** brackets is a resource and must contain `fxmanifest.lua` directly. Resource names must be unique across the whole server, wherever they sit.

Fusion backs this up in two ways (**Settings › Server › Resources**):

- 15 seconds after boot it starts anything in sub-folders of `[qb]` / `[addons]` that didn't start (switch this off, or list resources to leave stopped);
- folders the server can't see, like the two ✗ examples above, are listed in the server console when it starts, as *"<folder> does not have a resource manifest"*.

New Fusion modules start by themselves. A `fusion-…` module copied into `resources/[fusion]` (by hand, your host's file manager or an update) is started 20 seconds after boot, every time, so it works without touching `server.cfg`. Where the server lets Fusion edit `server.cfg`, it also adds the module's `ensure` line after the other Fusion lines (the old file is kept as `server.cfg.bak-fusion`); most servers don't allow that, and then the console shows the line once so you can add it yourself. `update-fusion` adds it too. Servers that start the whole folder with `ensure [fusion]` are left as they are. Turn this off in **Settings › Server › Resources**. To stop using a module, turn it off in **Settings › Modules** rather than deleting its line.

Vehicles, EUP, weapons, the carcols fixer and the handling editor all find packs in sub-folders automatically. **New vehicle packs and the carcols fixer / handling editor:** after adding a pack, restart the whole server once before fixing or editing it. FiveM only lets Fusion save into packs it knew about when the server started. Then fix, and restart again so players download the fixed pack. To install straight into a sub-folder: `install/update.ps1 -Mode script -Source pack.zip -Target "[addons]/[cars]"` (Linux: `bash install-script.sh pack.zip --target "[addons]/[cars]"`), or install normally and move the folder afterward: upgrades find it wherever it is.

Your own pictures and media stay yours: Fusion updates never overwrite them, and they are never part of a Fusion release.

9. Running QBCore scripts

Fusion provides the full QBCore API. The `[compat]` folder from the free server kit holds small bridge resources (`qb-core`, `qb-menu`, `qb-input`, `progressbar`) that pass QBCore calls on to Fusion, so scripts find the resource names they expect. Keep them in `resources/[fusion]/[compat]`.

- `exports['qb-core']:GetCoreObject()`, `QBCore.Functions.*`, the player object (`Player.Functions.AddMoney`, `SetJob`, `AddItem` ...), callbacks, commands, usable items, and `QBCore.Shared` items, jobs, gangs, vehicles and weapons.
- The standard `QBCore:*` events (player loaded/unloaded, job and duty updates, notifications...).
- QBCore's database tables: `players`, `player_vehicles`, `playerskins`, `player_outfits`, `bans`, `management_funds`, `inventories`.
- Exports from `qb-inventory`, `qb-target`, `qb-menu`, `qb-input`, `progressbar`, `qb-weathersync`, `qb-vehiclekeys`, `qb-management`, `qb-banking`, `LegacyFuel` / `cdn-fuel` / `ps-fuel`, and the common `hospital:*` and `prison:*` events.

Do not install these. Fusion already does their job, and installing them causes conflicts: `qb-core`, `qb-inventory`, `qb-target`, `qb-menu`, `qb-input`, `progressbar`, `qb-multicharacter`, `qb-spawn`, `qb-apartments` (spawn part), `qb-garages`, `qb-shops`, `qb-clothing`, `illenium-appearance`, `qb-vehiclekeys`, `qb-weathersync`, `qb-ambulancejob`, `qb-prison`, `qb-management`, `qb-banking`, `LegacyFuel`, `qb-hud` (money/fuel shown by Fusion).

Good to know:

- Scripts that use **PolyZone** directly need the PolyZone resource in `[qb]`.
- Items and jobs a QBCore script registers through `exports['qb-core']:AddItem` / `AddJob` are saved and then editable in the admin panel.
- QBCore permission levels map to the nodes `qbcore.mod`, `qbcore.admin` and `qbcore.god`.
- Most QBCore scripts work as-is, but a script that relies on the internals of a specific QB resource may need small fixes.

Script Configs (F10 › Script Configs)

Every script you add (anything outside `[fusion]`) is listed with its config files (`config.lua`, `shared/config.lua`, `config/*.lua`, `settings.json` and the like). Scripts new since the last start are marked **New**.

- **Settings:** every simple setting in the file (`Config.UseTarget = true`, `Config.Pay = 250`, `Config.Truck.Model = 'trash'` ...) is listed with the comment written next to it in the file. Change values with switches and boxes, then **Save changes**. Only the value changes; the rest of the line, comments and layout stay exactly as they were.
- **Edit as text:** the whole file in a large editor (the script list folds away while you edit; click **Settings** to bring it back), for lists, tables and anything else. A Lua file with an error (or broken JSON) is refused instead of saved, so a typo can't stop the script from loading.

- **Auto-configure for Fusion:** Fusion answers to the QBCore names, so most scripts just need pointing at them. Fusion suggests the values: framework → the QBCore word the script itself checks for (`qb` , `qbcore` ...), core resource → `qb-core` , inventory → `qb-inventory` / `qb` , target → `qb-target` (and *UseTarget* on), fuel → `LegacyFuel` , voice → `pma-voice` , keys → `qb-vehiclekeys` , notifications → `qb` , and so on. Uncheck anything you don't want and press **Apply selected**. Suggestions only appear for Fusion modules that are running.
- **Restart the script after saving** is checked for running scripts, so changes apply straight away. **Restart script** is also on the page.
- **Start with the server:** next to **Backups**, each script has one button. **Red** (*Add ensure (script name) to server config*): it doesn't start with the server yet; press it to add `ensure <script>` to `server.cfg` and start it now (the old file is kept as `server.cfg.bak-fusion`). **Green**: it already starts with the server, by its own line or with its whole folder (`ensure [qb]`); press it to turn its line into a `#` comment. Vehicle packs have no button (they start with their folder). Copied a new script onto the server? Press **Refresh** and it appears.
- **Start order:** Fusion reads what each script needs (its manifest's dependencies and the scripts it calls through `exports[...]`) and keeps your scripts' ensure lines in that order: a script starts after the scripts it needs. This happens when you add a line and at every server start; Fusion's own lines and folder lines don't move.
- **Start after...:** when a script has to wait for another one that Fusion can't see in its manifest or code, press **Start after...** on its page and pick the scripts it must start after. Fusion moves the ensure lines (or orders its own start list) straight away and keeps that order at every start. The button then shows what it starts after. If Fusion can't put them in order by itself, it says why and what to do: both scripts start with the same folder (FiveM picks the order inside a folder, so give the script its own ensure line), one of them doesn't start with the server, or the picks make a loop (A after B, B after A).
- **When server.cfg can't be written** from the server (FiveM blocks scripts from files outside the resources on many servers), the button still works: Fusion writes the scripts, in start order, as `ensure` lines into `resources/[fusion]/fusion-core/permissions/startup.cfg` , and server.cfg runs that file with one line at its very end: `exec resources/[fusion]/fusion-core/permissions/startup.cfg` (new installs and the updaters add it). Without that line Fusion starts them itself 20 seconds after boot, and the console shows the line to add.
- **Free script updates:** scripts published on GitHub are found by themselves (from their manifest, or the version check in their own code, e.g. ULC) and compared with their latest GitHub release at every start and every 6 hours. The script's page shows *GitHub owner/repo · you have 1.8.0 · latest v1.9.0* with **Update to ...**, **Check**, **Auto-update** (per script), **Undo update** and a pencil to set or fix the GitHub page; the list marks scripts with an update. **Check for updates** (top right) checks them all. Installing backs up the whole script to `fusion-core/script-backups` first, **keeps your edited config files** (the release's version is saved next to them as `config.lua.new` so you can compare), then restarts the script. Automatic installs happen at server start (before anyone plays) or later only while the server is empty; otherwise staff are told. Settings › Server › Resources: *Check free scripts for updates* and *Install free script updates automatically* (both on). Paid / escrowed scripts (`.fxap`), git checkouts and vehicle packs are never touched.
- **Backups / undo:** every save keeps a copy of the old file in `fusion-core/config-backups` first. Undo puts it back (and backs up the current file too). Backups are never part of a release or an update.

- **Auto-configure new scripts by themselves: Settings › Server › Resources › Auto-configure newly added scripts for Fusion** (off by default). When it's on, a script that wasn't there at the last start is configured for Fusion and restarted when the server boots, and each change is written to the console and the log. When it's off, the console just tells you a new script has settings to point at Fusion.

Like the handling editor, Fusion can only save into scripts the server knew about when it started: after adding a script, restart the server once before saving its config. Permission: `admin.scriptconfig`.

10. Keys and commands

Default keys

Every key below is a FiveM key binding: each player can change it in *GTA Settings › Key Bindings › FiveM* (look for names starting with **Fusion**). Prompts on screen always show the key the player chose.

KEY	ACTION	WHO
M	Main menu	Everyone
I	Inventory	Everyone
1 – 5	Use hotbar slot 1–5	Everyone
E	Interact: stores, garages, impounds, pumps, stashes, banks / ATMs, duty points, clothing stores, hospital check-in, respawn	Everyone
G	Second interaction: boss menu at a duty point, buy a jerry can	Everyone
X	Cancel the current action / progress bar	Everyone
L	Lock / unlock vehicle	Everyone
B	Seatbelt	Everyone
Left Alt (hold)	Third eye / target	Everyone
Left mouse	Pick the third-eye option you are looking at	Everyone
Caps Lock	Mouse cursor in the menu	Everyone
F1	Phone	Everyone
↑ ↓ ← →, **Enter**, **Backspace**	Menu: move, change value, select, back	Everyone
A / D (hold)	Turn your character in the appearance editor	Everyone
TAB	Fusion weapon wheel (when *Replace GTA's weapon wheel* is on)	Everyone
Q / R / Y, hold **E**	Emergency lights / siren / next tone / air horn (in emergency vehicles)	Per job / permission
F5	Police radar power	Per job / permission
Numpad 8 / 5	Lock front / rear radar speed	Per job / permission
Numpad 9 / 6	Lock front / rear radar plate	Per job / permission

Numpad 0	Start / reset the PIT timer	Per job / permission
F6	Siren controller / radar remote with the mouse	Per job / permission
F11	Voice range (whisper / normal / shout)	Everyone
Caps Lock (hold)	Talk on the radio	Everyone with a radio
F10	Admin panel	Staff
Home	Noclip on / off	Staff
Q / E, **Left Shift / Left Ctrl**	Noclip up / down, faster / slower	Staff
Backspace	Stop spectating	Staff
Delete	Delete the entity under the entity laser. A prop of GTA's map stays deleted for everyone, also after restarts (`/restoreprops` brings them all back); a prop built with the Interior Builder is deleted from its place. Props built into GTA's interiors can't be hidden this way	Staff
E / Q, **Left Shift / Left Ctrl**, mouse wheel, **Z / C**, **X**, **Backspace**	Free camera: up / down, faster / slower, zoom, tilt, level, leave	Players with `media.freecam`
Controller: your own buttons (*Fusion: … (controller, set your button)*, no default)	Emergency lights / siren / next tone / air horn	Per job / permission

Fusion add-ons work the same way: the **Fusion Interior Builder**'s build-mode keys are listed as *Fusion Builder: …*.

Three things follow GTA's own controls instead (change them in GTA's normal key bindings): walking in noclip and the free camera (your walking keys), **R** to reload (GTA's *Reload* key; with an empty weapon it loads ammo from your inventory), **Esc** to close the menu, and the controller D-pad in menus.

In the menu: ↑ ↓ move, ← → change values, **Enter** select, **Backspace** back, **Esc** close. Players can keep walking and driving while it's open.

Commands

COMMAND	WHO	WHAT
`/admin`	Staff	Admin panel
`/guide`	Everyone	Reopen the player guide
`/hud`	Everyone	Move or hide HUD elements
`/calladmin [reason]`, `/report <message>`	Everyone	Ask staff for help

`/me <action>`, `/does <description>`	Everyone	Roleplay: shown above your character for 7 seconds and in the chat of players within 20 m (**Settings › Gameplay › Roleplay**)
`/reports`	Staff	Open the Reports tab
`/noclip`	Staff	Toggle noclip
`/car <model>`, `/dv`, `/fix`	Per permission	Spawn, delete or repair a vehicle
`/wheel`	Everyone	Weapon wheel
`/duty`, `/job`, `/cash`, `/bank`	Everyone	Duty and info
`/logout`	Everyone	Back to character selection
`/kick`, `/ban <id> <hours> <reason>`, `/warn`	Staff	Moderation (0 hours = permanent)
`/revive [id]`, `/heal [id]`, `/goto`, `/bring`, `/freeze`, `/kill`	Staff	Player actions
`/setjob <id> <job> <grade>`, `/setgang`	Staff	Jobs
`/givemoney <id> <account> <amount>`, `/setmoney`, `/giveitem <id> <item> <amount>`	Staff	Money and items
`/setrole <id> <role>`, `/removerole`	Staff	Roles
`/announce <message>`	Staff	Server announcement
`/jail <id> <minutes> <charges>`, `/unjail <id>`	Police / staff	Jail
`/weather <type>`, `/time <h> <m>`, `/blackout`	Staff	Weather and time
`/weatherevent thunderstorm|flood|hurricane|tornado|blizzard|hailstorm|stop`	Staff	Severe weather events
`/setfuel <0-100>`	Staff	Set your vehicle's fuel
`/postal [code]`	Everyone	Waypoint to a postal code (empty clears it)
`/streamer`, `/videomode`, `/freecam`	Everyone	Streamer mode, video mode, free camera
`/record`, `/clips`	Everyone	Start / stop recording, open My Clips
`/claimowner <code>`	Anyone (once)	Become owner of a server that has none (code is in the server console)

`fusion_role <id	identifier> <role>`	Server console

11. Backups and updates

- **Settings backup:** F10 › Settings › **Export** copies every setting as text. **Import** restores it.
- **Full backup:** back up the `fusion` database (phpMyAdmin › Export, or `mysqldump fusion > fusion.sql` on Linux) and the server folder. Everything Fusion knows lives in those two places.

Automatic updates (F10 › Updates)

Fusion checks for new versions by itself (every 6 hours) and tells staff with `admin.updates` (owners by default) when one is out.

- **F10 › Updates** shows the version you run, the newest one, and **what changed** in every version in between.
- New versions are **downloaded by themselves** and checked before anything can be installed: the update list must carry Fusion Platform Technologies' signature (a fake or altered list is ignored), and the download must match its size and **SHA-256 checksum**, and every file inside it its own checksum. A broken or tampered download is thrown away and downloaded again. If the main download address fails, it retries, resumes where it stopped, and moves on to the next mirror.
- **You choose when it installs: Install** (players get a countdown, by default 5 minutes, then everyone is disconnected while Fusion restarts, about a minute), **When the server is empty**, or **At this time**. Or let it install by itself in Settings › Updates (when the server is empty, or at a set time each day). **Skip this version** if you want to wait for the next one.
- **Installing:** Fusion stops its modules, backs up exactly the files it is about to change, writes the new ones and reads every one back to check it. Files you edited are kept (the new version goes beside them as `.new`), and your pictures, media, clips and settings are never touched, the same as the updater script.
- **If anything goes wrong it puts the old version back by itself:** when a file can't be written, when the new version doesn't start and report its build within 3 minutes, or when the server stops in the middle of an install (the next start restores the old files before Fusion loads). Staff are told why, and that build isn't installed by itself again.
- **Undo** in the history list goes back to the version you had before the last update.
- **Fusion add-ons update too.** Every Fusion add-on installed here (Trucking, Fires, Aerial, Lighting, Interior Builder, Pursuit Equipment, Fire Hose) is listed under *Add-ons* with its version and the newest one. New versions are checked and downloaded the same way, and installed the way you set in Settings › Updates (or with **Install add-on updates**). Each add-on is backed up and restarted on its own for a few seconds while players stay connected; if it doesn't start, its old files go back by themselves and that version isn't installed by itself again. Files you edited are kept, with the new version saved next to them as `.new`. Only copies from Cfx.re (escrowed) are updated. Servers that run an add-on without Fusion Menu are updated by the add-on's own updater, which stays off while `fusion-updater` runs.

The updater is its own resource, `fusion-updater`. It must start **before** `fusion-core` (`ensure fusion-updater` above `ensure fusion-core`; new installs and updates put it there), so it can repair Fusion even when Fusion itself doesn't start. It saves its downloads and backups in `resources/[fusion]/fusion-updater/data` (the last 3 versions by default, Settings › Updates).

A server that runs straight from Fusion's source code (a development copy) only shows updates and never installs them.

Updating Fusion Menu with the updater script

1. Download the new version from your **Cfx.re keymaster** (Granted Assets) and put the zip in the `updates` folder.
2. Stop the server (txAdmin › Stop server, or `sudo systemctl stop fusion`).
3. Run `update-fusion.bat` (Windows) or `bash update-fusion.sh` (Linux).
4. Start the server.

The updater:

- backs up every Fusion resource, your `server.cfg` and the **database** to `backups/<date>/` first (the database backup needs `mysqldump`, which XAMPP and MariaDB include);
- replaces the framework files, removes files the new version no longer uses, and installs new modules (adding any missing `ensure fusion-…` line to `server.cfg`, in order after your other Fusion lines);
- **keeps everything you changed or added**: logos, loading screen media, item / vehicle / weapon / uniform pictures, and any file you edited;
- refuses to install an older version over a newer one (add `-Force` / `--force` if you really mean to).

Your settings, roles, jobs, items, characters and vehicles live in the database and are untouched; new settings simply appear with their defaults. Never delete the `[fusion]` folders and put new ones in their place: that throws away your pictures and media. Copying the new files *over* them is fine (next section).

Updating by hand (game server hosts, FTP, file manager)

For servers where you can't run the updater (ZAP-Hosting and other txAdmin hosts, or if you prefer to do it yourself):

1. Back up first: F10 › Settings › **Export**, the database (your host's panel or phpMyAdmin › Export), and download a copy of `resources/[fusion]`.
2. Stop the server.
3. Unzip the new Fusion zip on your PC.
4. Upload every `fusion-…` folder from the zip over the folder of the same name in `resources/[fusion]`, **replacing files but not deleting the old folders first**. Leave `resources/[fusion]/[compat]` as it is: those bridges come with the free server kit, not the Fusion zip. Never end up with two copies of the same folder name anywhere in `resources`: the server refuses to start a resource it finds twice.
5. **Don't** upload over or delete: `server.cfg`, `resources/[qb]`, `resources/[addons]`, `resources/[standalone]`, and inside `fusion-core` the folders `support`, `permissions`, `media-clips`, `config-backups` and `script-backups`. If you replaced a built-in picture (an item, vehicle or weapon picture with the same name as Fusion's), skip that one file when your FTP program asks.
6. Start the server.

After updating

- The server console prints the version and build number at start (`Fusion Menu v1.0.0 build …`), and the loading screen and F10 show it too. Check it's the new one.
- New Fusion modules start by themselves about 20 seconds after start. `update-fusion` also adds their `ensure` line to `server.cfg` ; hosted servers can add it by hand or leave it (Fusion keeps starting them).
- New settings appear with their defaults in F10 › Settings. Read the release notes for anything worth switching on.
- **Linux:** if you uploaded the files as a different user than the one the server runs as (e.g. SFTP as `ubuntu` , or `root`), give them back to the server's user so Fusion can save files: `sudo chown -R <user>:<user> /home/fivem/fusion` (the user the `fusion` service runs as, normally the one who ran the installer).
- **Clips as MP4:** the Fusion Recorder needs ffmpeg on the server once (`winget install ffmpeg` on Windows, `sudo apt install -y ffmpeg` on Linux). Without it clips stay WebM.

Updating the installer tools

The updater and script installer (`update-fusion.*` , `update-fxserver.*` , `install-script.*` , `install/`) come from the free server kit. To get their newest version, download the kit zip from https://github.com/Marine1108/fusion-menu-server-kit/releases/latest and copy `install/` , `update-fusion.bat` , `update-fusion.sh` , `update-fxserver.bat` , `update-fxserver.sh` , `install-script.bat` and `install-script.sh` over your server folder. Nothing else in the kit needs copying on an existing server.

Updating FiveM and txAdmin (FXServer)

txAdmin is part of the FiveM server program (**FXServer**, also called the *artifacts*), so updating txAdmin means updating FXServer. When txAdmin says *"A patch update is available for txAdmin"*, this is what to do. It doesn't touch Fusion Menu, your scripts, the database, `server.cfg` or txAdmin's settings (`txData`).

FiveM offers two builds:

- **recommended**: tested by FiveM; the safe choice and what Fusion's installers use.
- **latest**: the newest build, with the newest txAdmin patch. A brand-new txAdmin patch usually reaches *latest* days or weeks before *recommended*.

Your txAdmin panel shows its version at the bottom of the menu, and the console prints the FXServer build when it starts.

Windows (installed with Fusion's installer)

1. Stop the server: txAdmin › **Stop Server**, or close the server window.
2. In your server folder, double-click `update-fxserver.bat` . For the newest txAdmin patch, open a command prompt in the server folder and run `update-fxserver.bat latest` .
3. When it says *FXServer updated to build …*, start the server again with the **Fusion txAdmin** (or **Start Fusion Server**) shortcut.

It finds FXServer through the start shortcuts (normally `C:\FXServer`). If yours is somewhere else: `set FX_DIR=D:\path\to\FXServer` then `update-fxserver.bat` .

Linux VPS (installed with `install-linux.sh`)

Over SSH, in your server folder (e.g. `/home/fivem/fusion`):

```
sudo bash update-fxserver.sh          # recommended build
sudo bash update-fxserver.sh latest   # newest build (newest txAdmin patch)
```

It stops the `fusion` service, swaps in the new FXServer, keeps txAdmin's `txData` where it is, and starts the service again. It finds FXServer through the `fusion` service (normally `/opt/fxserver`). If yours is somewhere else: `sudo FX_DIR=/path/to/fxserver bash update-fxserver.sh`.

Undo (Windows and Linux): the previous build is kept until the next update. If the new one misbehaves, `update-fxserver.bat rollback` or `sudo bash update-fxserver.sh rollback` puts it back.

Game server hosts (ZAP-Hosting and others)

The host installs FXServer for you, so it's updated in their panel, not with files:

1. Stop the server in the host's panel.
2. Find the FiveM version setting. Hosts call it *Version*, *Server version*, *Artifact* or *FXServer build*, usually under *Settings* or the game server's *Versions* page.
3. Choose **Recommended**, or **Latest** / the highest number for the newest txAdmin patch, and save. Some hosts reinstall the server program at this point; your files and database stay.
4. Start the server.

If you can't find the setting, ask the host's support to *"update FXServer artifacts to the latest build"*.

Installed FXServer yourself (txAdmin recipe on your own PC or VPS)

1. Stop the server.
2. Download the build from FiveM: Windows (`server.zip`) or Linux (`fx.tar.xz`). The page marks the **recommended** and **latest** builds at the top.
3. Unpack it over your FXServer folder, replacing the old files. **Don't delete the folder first** if txAdmin's `txData` is inside it.
4. Start the server.

Upgrading QBCore scripts and packs

Drag the new version onto `install-script.bat` (or `bash install-script.sh new-version.zip`). Files you changed (`config.lua`, locales, anything else) are kept, and the script's new version of each one is saved next to it as `config.lua.new`. Open both, copy over any new options you want, then delete the `.new` file.

The very first upgrade of a script you installed by hand keeps its config and locale files (`config*.lua`, `config/`, `locales/`, `.cfg`); every install after that tracks exactly which files you edited.

How it knows what you changed

Each install writes a `.fusion-install.sha256` file into the resource folder: a fingerprint of every file it put there. On the next update, a file that still matches its fingerprint is replaced; a file that doesn't (you edited it) is kept. Leave this file in place. Every run also writes `report.txt` into its backup folder.

To roll back, stop the server and copy the resource folders from `backups/<date>/` back into `resources` (restore `database.sql` with phpMyAdmin › Import if needed).

12. License

Fusion Menu (the `fusion-…` resources) is licensed, not sold (see `LICENSE`). You may use and modify it on servers you own or operate. You may **not** resell, share or redistribute it, original or modified, and the *Fusion Menu by Fusion Platform Technologies* credit must stay. The free server kit (installers, setup tools, the `server.cfg` template and the `[compat]` bridges) may be used, changed and shared, but not sold. oxmysql (LGPL-3.0), the default Cfx.re resources (MIT) and the built-in postal code sets from nearest-postal (MIT) keep their own licenses. QBCore and the other projects named here belong to their owners; Fusion Menu is compatible with them and not affiliated with them.

13. Troubleshooting

Asking for help: open a ticket on our Discord (https://discord.gg/kVRqwDPsnJ) and attach a support file:

1. **F10 › Support › Create support file**, or type `fusionsupport` in the server console if F10 won't open.
2. It's saved as `resources/[fusion]/fusion-core/support/fusion-support—<date>.txt`. F10 also has **Copy everything**.

It holds your Fusion build number (also shown at the bottom right of the loading screen), the FXServer and txAdmin versions, every resource and whether it started, the latest Fusion log and the latest 2500 lines of the server console. Passwords, license keys, database logins, webhooks and players' IP addresses are blanked out.

PROBLEM	FIX
Console: *You lack the required entitlement* for a fusion resource	The server's license key belongs to a different Cfx.re account than the one that bought Fusion Menu. Use a key from the buying account
Console: *Still waiting for the database*	MySQL isn't running, or the database doesn't exist. Start MySQL (XAMPP) or `systemctl start mariadb` (Linux) and run the installer again
Console runs old resources (another framework, missing cars…)	txAdmin's *CFG File* still points at another server's `server.cfg`
Stuck on a camera view or blank screen	Press F8 and type `nui_devtools` to see interface errors; send them to whoever maintains your server
A picture doesn't show	Check the file name matches exactly (including `.png`) and `ensure` the resource
A third-party script ignores your roles	Find the permission name it uses and add it under **Custom nodes** (section 5)
A QBCore script can't find `qb-xxx`	Check it isn't on the *do not install* list. If it needs PolyZone, add PolyZone to `[qb]`
Locked out of the admin panel	Run `fusion_role <your server id> superadmin` in the server console
Linux: resources or pictures not found	File names are case-sensitive on Linux. `Car.png` and `car.png` are different files
VPS: players can't connect	Open port 30120 TCP and UDP in your provider's firewall as well as `ufw`
An updated script misbehaves	Look for `.new` files in its folder: the new version may need options your kept `config.lua` doesn't have. Or roll back from `backups/<date>/`
The updater says the server is running	Stop it in txAdmin (or `sudo systemctl stop fusion`) and run it again

F10 › Updates says *permission denied*, *FiveM refused the write* or *FiveM does not let the updater write*	The updater may only write Fusion's folders when `server.cfg` runs `exec resources/[fusion]/fusion-core/permissions/filesystem.cfg` **above** `ensure fusion-updater`, and the folders belong to the user the server runs as: on Linux `sudo chown -R <user>:<user> /home/fivem/fusion` (the user in `User=` of `systemctl cat fusion`). Restart, then install again
Carcols fixer or handling editor says FiveM refused to edit a pack	Restart the server once (new packs need one restart). If it repeats, check `server.cfg` contains `exec resources/[fusion]/fusion-core/permissions/filesystem.cfg`
Players spawn in the wrong place	Check **Settings › Spawning** (spawn mode and locations)
A legit player was kicked or banned by the anticheat	Look at their row in **F10 › Anticheat** to see why. Unban them in **Bans**, then set that detection to *Log only*, raise its threshold, or give them `anticheat.bypass`

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